Development of Machine Learning Algorithms for Automatic Geotag Recognition in Textual Data

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Abstract

Geographic information extraction from unstructured textual data has become increasingly critical in domains ranging from smart city management to emergency response systems. This thesis presents a comparative study of machine learning approaches for automatic geotag recognition — a form of Named Entity Recognition (NER) — applied to a real-world multilingual dataset of public transport complaint records in Kazakh and Russian languages, provided by City Transportation Systems (CTS) of Astana, Kazakhstan.

Four distinct approaches are developed and evaluated: (1) fine-tuned XLM-RoBERTa, (2) fine-tuned ruBERT, (3) a few-shot Large Language Model pipeline based on Qwen2.5 (7B) deployed locally via Ollama, and (4) a fine-tuned Multilingual-E5-base (Me5-base) encoder. The dataset comprises 2,750 manually annotated complaint records with six domain-specific entity types: `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION`, annotated using the BIO tagging scheme.

Experimental results demonstrate that the Qwen2.5 LLM achieves the highest overall precision (0.788) and macro F1-score (0.635), and is the only model capable of detecting the complex `STREET_INTERSECTION` entity. Me5-base achieves the best recall (0.653). XLM-RoBERTa provides the strongest single-entity performance on `ROUTE_NUM` (F1 = 0.920), while ruBERT (F1 = 0.223) is shown to be inadequate for code-switching multilingual data. These findings provide practical guidance for deploying geotag recognition systems in low-resource, multilingual operational environments.

Keywords: Named Entity Recognition, Geotag Extraction, XLM-RoBERTa, BERT, Large Language Models, Multilingual NLP, Kazakh, Russian, Transport Complaints, BIO Tagging, Sequence Labeling

Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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Introduction

The accelerating digitisation of urban transport infrastructure has generated an ever-growing volume of unstructured citizen feedback. Public transport authorities now receive thousands of complaint messages daily through mobile applications, web portals, and call-centre transcriptions. Each record contains a free-form description in which passengers report operational failures: missed stops, route deviations, driver misconduct, and payment issues. Embedded within this unstructured text are geographic references — route numbers, stop names, street names, vehicle identifiers, and intersection descriptions — that, if extracted automatically, can power real-time situational awareness dashboards, smart route-optimisation engines, and automated incident-routing systems. The growing volume of these records, combined with the operational cost of manual triage, highlights a critical need for effective and broadly available automated text analysis capabilities. Meanwhile, recent advances in natural language processing and transformer-based language models have made it possible to detect semantic patterns in noisy multilingual text, rendering automatic geotag extraction an immediate and realistic prospect.

Despite the expanding body of research on named entity recognition, most existing systems have been developed for well-resourced languages and generic domains such as news articles or biomedical literature. These systems lack the specificity needed for transport-domain operational text in low-resource bilingual settings. In the context of Astana, Kazakhstan, where City Transportation Systems (CTS) processes approximately 3 000 complaint records per month, the target text is written in a mixture of Kazakh and Russian with frequent code-switching, informal register, domain-specific abbreviations, and inconsistent punctuation. No prior work targets the specific combination of Kazakh-Russian code-switching, transport-domain complaint text, a six-class entity taxonomy including structurally complex composite entities, and a comparative evaluation of architecturally distinct machine learning approaches under identical experimental conditions. This gap represents a substantial drawback, as no appropriate tool currently exists to support automated complaint triage for a multilingual Central Asian public transport operator.

The current study aims to address this gap by developing and comparing multiple machine learning approaches for automatic geotag recognition in multilingual textual

data, motivated by the operational needs of CTS.

To realise this goal, the study (i) reviews the state of the art in named entity recognition and geotag extraction across rule-based, statistical, neural, and large language model paradigms; (ii) collects, preprocesses, and annotates a domain-specific multilingual dataset of 2 750 public transport complaint records with gold-standard BIO labels for six entity types; (iii) develops and evaluates four machine learning approaches — fine-tuned XLM-RoBERTa, fine-tuned DeepPavlov ruBERT, fine-tuned Multilingual-E5-base, and few-shot Qwen2.5 LLM inference — together with four baseline methods against per-class and macro-averaged precision, recall, and F1 metrics; and (iv) integrates the best-performing architecture into a recommended inference pipeline suitable for real-time CTS complaint processing. The research object is the process of named entity recognition in multilingual public transport complaint texts; the research subject is the family of machine learning models and algorithms for automatic extraction of six domain-specific geotag entity types from noisy, code-switched Kazakh-Russian natural language text.

The dissertation is composed of four chapters. Chapter 1 outlines the theoretical foundations by analysing the linguistic and operational characteristics of geographic named entities in transport-domain text. It reviews current approaches to automatic named entity recognition from rule-based and statistical methods through transformer encoders and large language models, examines the specific challenges of Kazakh-Russian code-switching and morphological complexity, and summarises 21 key related works. On this basis, the chapter formulates the research problem, defines the six entity types, specifies the data requirements for model training, and establishes the quantitative criteria for evaluating model effectiveness.

Chapter 2 addresses model development and experimental setup, presenting the data collection and annotation pipeline, the preprocessing methodology for handling code-switching and subword alignment, the training configurations for each of the four main models, the design of the four baseline methods, and the evaluation metrics used throughout the study.

Chapter 3 presents the comparative experimental results, including per-model performance tables, error analysis of challenging cases — rare entities, code-switching failures, and noisy input — and a cross-model comparison leading to a deployment recommendation for the CTS production environment.

Chapter 4 summarises the principal findings, states the research contributions, and outlines directions for future work including targeted data augmentation for rare entities

and hybrid architectures combining encoder and LLM components.

The applied value of this work lies in its direct relevance to real-world public transport management. By enabling rapid and accurate geotag extraction from citizen complaints, the proposed system allows transport operators to respond to incidents faster, identify geographic hotspots automatically, and reduce dependence on manual triage operators. In addition, the annotated dataset and comparative results provide a baseline for subsequent research at the intersection of multilingual NLP and smart city infrastructure.

Chapter 1

Theoretical foundations and domain analysis

1.1 The relevance of automatic geotag recognition in textual data

The accelerating digitisation of urban services has produced an ever-growing stream of unstructured operational text. Public transport operators, emergency services, and municipal authorities now receive thousands of complaint messages, incident reports, and citizen feedback submissions every day. Buried within these free-form texts are geographic references — street names, transit stop names, route numbers, and intersection descriptions — that carry high operational value if extracted reliably and in real time. The challenge of identifying and structuring these geographic references from noisy, multilingual user-generated content has become one of the central problems at the intersection of natural language processing and geospatial computing.

The broader context is the global Smart City movement. Smart City frameworks aim to improve urban life quality by integrating data-driven information and communication technologies into core municipal functions. Transport management is among the highest-impact application domains: transit authorities must maintain fleet reliability, respond rapidly to operational failures, and continuously adapt routes to shifting passenger demand. Traditional management approaches rely on structured telemetry — GPS positions, passenger counts, schedule adherence — but such data captures *where* and *when* events occur without explaining *why*. Citizen complaints fill this explanatory gap. When a passenger reports that «автобус маршрут 48 не остановился на Байтурсынова» ("bus route 48 did not stop at Baitursunov"), they encode actionable intelligence: the route identifier, the missed stop name, and an implicit geolocation that, if extracted

and linked to the city's GIS, can trigger a dispatcher alert within seconds.

City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan, provided the concrete motivation for this work. CTS processes approximately 3 000 complaint records per month received through a mobile application, a web portal, and a call-centre. Each complaint contains a free-text description field (*описание*) in which passengers report transportation issues such as missed stops, route deviations, driver misconduct, and payment failures. The current triage workflow requires a human operator to read each complaint, identify the relevant vehicle, stop, and street, and manually route the record to the responsible dispatcher. This process is labour-intensive: a single operator can process at most 60–80 records per hour, meaning that during high-volume periods, complaints may wait two to four hours before reaching the appropriate authority.

Automation of geotag extraction directly addresses this bottleneck. If a reliable NLP model can identify the vehicle identifier, the stop name, and the street reference in each complaint, the triage and routing steps can be fully automated. Moreover, extracted geotags enable downstream analytical applications of considerable value:

- **Hotspot detection.** Geographic clustering of complaints reveals stops or intersections with disproportionately high incident rates, enabling proactive maintenance and rescheduling.
- **Route-level trend analysis.** Aggregating route identifiers over time detects systematic reliability degradation before it becomes visible in aggregate metrics.
- **Dispatcher workload balancing.** Associating complaints with geographic zones enables more even distribution of operator workload.
- **GIS overlay and visualisation.** Extracted geotags, resolved against the city's stop and street databases, populate real-time situational awareness dashboards used by transport management.
- **Policy evaluation.** Before-and-after analysis of complaint geotag distributions can evaluate the impact of route changes, infrastructure investments, or driver training programmes.

The problem is non-trivial because the CTS corpus exhibits several characteristics that challenge standard NLP pipelines. First, complaint texts are written in a mixture of Kazakh and Russian, with code-switching occurring both between and within

sentences. Second, the texts adopt an informal register with abbreviations («м/а» for «маршрут / автобус»), non-standard punctuation, spelling errors, and colloquial expressions. Third, the six geographic entity types targeted by this work differ substantially in their linguistic structure, frequency, and extraction difficulty, creating a heterogeneous NER task in which a single modelling strategy cannot optimally serve all entity types simultaneously. Fourth, the dataset is severely class-imbalanced: the most common entity type (`STOP_NAME`) occurs more than 10 times more frequently than the rarest (`STREET_INTERSECTION`), making standard maximum-likelihood training suboptimal for minority classes.

Named Entity Recognition (NER) is the natural language processing task most directly applicable to this problem. NER frames geotag extraction as a supervised sequence labelling problem in which each token in the input is assigned a semantic label denoting its entity class and its position within an entity span. The accuracy, robustness, and computational efficiency of the chosen NER architecture directly determine the practical viability of an automated geotag recognition pipeline. This thesis develops and compares four fundamentally different approaches — fine-tuned multilingual transformer encoders, a Russian-specific BERT model, a multilingual embedding-based model, and a large language model deployed via few-shot prompting — to identify which strategy or combination of strategies best meets the operational requirements of the CTS geotag extraction task.

The research builds upon a related prior study [1] that established the heterogeneous nature of transport-domain NER and demonstrated that aggregate evaluation metrics can conceal important per-category differences. The present thesis extends that work with a comprehensive multi-model comparison and provides practical guidance for selecting and deploying NER models in similar low-resource, bilingual operational environments.

1.2 General characteristics of geographic named entities and toponyms in transport-domain texts

Geographic Named Entities (GNEs) in natural language span a wide spectrum of granularity: from broad administrative units (countries, regions, cities) through intermediate structures (districts, postal zones) to fine-grained operational objects (individual streets, building addresses, transit stop names). In the transport complaint domain, the operationally relevant GNEs are uniformly fine-grained and domain-specific, differing substan-

tially from the entity types targeted by standard NER benchmarks such as CoNLL-2003, which groups all spatial references into a single `LOC` class regardless of type or function.

Geotags in public transport complaints serve three distinct operational functions. *Identification tags* (route numbers, vehicle identifiers, licence plates) link a complaint to a specific operational asset that can be located in the fleet management database. *Spatial tags* (stop names, street names) link a complaint to a geographic location that can be resolved against the city's GIS stop and street layer. *Composite tags* (street intersections) provide higher-resolution spatial references by combining two or more spatial references within a structured expression. Each functional class has different structural properties, annotation requirements, and extraction difficulty levels.

Based on corpus analysis and domain requirements, six entity types were defined for the CTS geotag recognition task:

`ROUTE_NUM`	Public transit route number. Examples: «Маршрут 47», «автобус № 64», «48 бағыт». These identifiers typically appear as short numeric spans, optionally preceded by a route keyword.
`BUS_ID`	Vehicle identifier or fleet (garage) number assigned to a specific bus. Examples: «гаражный номер 3056», «борт 10815». Unlike route numbers, bus IDs identify the physical vehicle rather than the service.
`PLATE_NUM`	Vehicle licence plate number. Examples: «КЛС 012», «164 BW 01». Plates may combine Cyrillic and Latin characters in various regional format conventions.
`STOP_NAME`	Named transit or bus stop. Examples: «аялдама Қобыланды», «остановка Сарыарка», «ТРЦ Керуен». Stop names are the most frequent entity type and appear in both Kazakh and Russian forms.
`STREET_NAME`	Named street, avenue, boulevard, or road. Examples: «проспект Тауелсиздик», «улица Абая», «Кабанбай батыра даңғылы». Street names are morphologically inflected in context.
`STREET_INTERSECTION`	Geographic intersection of two or more streets, expressed as a structured multi-word span. Examples: «угол Абая и Сейфул-

лина», «пересечение Тауелсиздик и Республики». This is the rarest and structurally most complex entity type.

Table 1.1 summarises the structural characteristics of the six entity categories, including their linguistic nature, typical span length in tokens, and the principal sources of extraction difficulty.

Table 1.1: Structural characteristics of entity categories in the CTS corpus (adapted from [1]).

Entity Type	**Nature**	**Typical Span (tokens)**	**Principal Extraction Difficulty**
`ROUTE_NUM`	Numeric / identifier	1–4	Boundary ambiguity with time codes and QR identifiers
`BUS_ID`	Numeric / identifier	1–3	Semantic confusion with `PLATE_NUM`
`PLATE_NUM`	Alphanumeric	1–3	Mixed Latin-Cyrillic format variation
`STOP_NAME`	Nominal / toponym	1–8	Abbreviation, bilingual lexical variation
`STREET_NAME`	Morphological	2–6	Case inflection, code-switching boundary
`STREET_INTERSECTION`	Composite / structured	5–12	Internal compositional structure; rare training examples

`ROUTE_NUM` and `STOP_NAME` are the most frequent entity types, appearing in approximately 72 % and 68 % of complaint records respectively. `STREET_INTERSECTION` is the rarest, appearing in fewer than 3 % of records, and it is structurally the most complex: it spans 5–12 tokens and includes conjunction words (*угол*, *пересечение*, *қиылысы*), two distinct street names in inflected form, and optional directional or temporal qualifiers.

Transport-domain toponyms differ from general-language toponyms in several ways that complicate automatic extraction. *Context dependence:* the same string, such as «Республика», may refer to a bus stop (`STOP_NAME`), a street (`STREET_NAME`), or an organisation depending on surrounding context. *Abbreviation:* stop names and street types are frequently abbreviated in complaint text («ул.» for «улица», «пр.» for «проспект»,

«м/а» for «маршрут/автобус»). *Bilingual surface forms:* the same stop may appear as «Байтурсынова» (Russian) or «Байтурсынов даңғылы» (Kazakh) within the same document collection, requiring a model to generalise across orthographic variants. *Identifier confusion:* BUS_ID and PLATE_NUM both appear as short alphanumeric strings with limited surrounding context, leading to frequent inter-class confusion in models that rely primarily on surface form.

The entity distribution is severely skewed. Dataset analysis reveals that the `O` (outside) class accounts for over 85 % of all tokens, with entity tokens representing less than 15 % of the corpus. Among entity tokens, `STOP_NAME` and `ROUTE_NUM` account for approximately 65 % of all entity occurrences, while `STREET_INTERSECTION` represents under 1 %. This extreme imbalance necessitates class-weighted training objectives, oversampling strategies, and macro-averaged evaluation metrics that give equal weight to all entity categories regardless of frequency.

1.3 Modern approaches to automatic named entity recognition

The field of Named Entity Recognition has evolved through three major paradigms over the past three decades: rule-based and statistical methods, neural sequence models, and pre-trained transformer architectures. Each paradigm offers different trade-offs between linguistic coverage, labelled data requirements, computational cost, and performance on low-resource or code-switching settings.

1.3.1 Rule-based and statistical methods

The earliest NER systems relied entirely on handcrafted rules. Dictionaries of known named entities (gazetteers), regular expression patterns matching orthographic conventions (capitalisation, punctuation, digit sequences), and manually encoded linguistic context rules formed the backbone of systems such as ANNIE and LingPipe. For highly structured identifiers — licence plates, route numbers, postal codes — regex-based approaches achieve near-perfect precision because the surface form is tightly constrained by a well-defined format. However, recall suffers sharply for any variant not explicitly encoded in the pattern set.

Hidden Markov Models (HMM) extended rule-based systems with probabilistic sequence modelling. In an HMM-based NER system, the observed output is the token

sequence and the hidden states are the BIO labels. The model maximises the joint probability:

$$P(\mathbf{y}, \mathbf{x}) = \prod_{i=1}^{n} P(y_i \mid y_{i-1}) \cdot P(x_i \mid y_i) \tag{1.1}$$

where $P(y_i \mid y_{i-1})$ is the transition probability between adjacent labels and $P(x_i \mid y_i)$ is the emission probability of token x_i given label y_i. Optimal label sequences are decoded using the Viterbi algorithm. HMMs capture first-order label dependencies but cannot model long-range context or arbitrary feature functions.

Conditional Random Fields (CRF) [2] addressed the primary limitation of HMMs by modelling the *conditional* probability $P(\mathbf{y} \mid \mathbf{x})$ rather than the joint. A linear-chain CRF defines:

$$P(\mathbf{y} \mid \mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp\left(\sum_{i=1}^{n} \sum_{k} \lambda_k f_k(y_{i-1}, y_i, \mathbf{x}, i) \right) \tag{1.2}$$

where f_k are feature functions, λ_k are learned weights, and $Z(\mathbf{x})$ is the partition function ensuring normalisation. The feature functions can capture arbitrary properties of the input sequence (prefix/suffix patterns, capitalisation, dictionary membership) without the independence assumptions imposed by generative models. CRF-based systems with rich feature engineering held state-of-the-art NER performance for nearly a decade.

BiLSTM-CRF represents the natural bridge between classical statistical methods and modern neural approaches. A bidirectional Long Short-Term Memory network encodes each token using both left-to-right and right-to-left context:

$$\overrightarrow{\mathbf{h}}_i = \mathrm{LSTM}_{\mathrm{fwd}}(e_i, \overrightarrow{\mathbf{h}}_{i-1}), \quad \overleftarrow{\mathbf{h}}_i = \mathrm{LSTM}_{\mathrm{bwd}}(e_i, \overleftarrow{\mathbf{h}}_{i+1}) \tag{1.3}$$

$$\mathbf{h}_i = [\overrightarrow{\mathbf{h}}_i; \overleftarrow{\mathbf{h}}_i] \tag{1.4}$$

The concatenated hidden state $\mathbf{h}_i$ is passed to a linear layer producing label logits, which are then decoded jointly across the entire sequence by the CRF layer. The CRF enforces globally consistent label sequences (e.g., an I- tag must follow a B- tag of the same type). BiLSTM-CRF achieved state-of-the-art NER on CoNLL-2003 before transformer models superseded it and remains a competitive baseline for low-resource settings.

The key limitation of rule-based and statistical methods is their reliance on hand-crafted features that do not generalise to morphologically rich or low-resource languages. For Kazakh-Russian code-switching text, feature engineering becomes prohibitively expensive because the same entity type may surface in dozens of orthographic forms across two languages with complex and asymmetric morphological systems.

1.3.2 Transformer-based encoders (BERT, RoBERTa, XLM-R)

The transformer architecture [3] introduced the self-attention mechanism, which computes contextual representations by attending over all positions in the input simultaneously. For a query matrix Q, key matrix K, and value matrix V derived from the input:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{1.5}$$

where d_k is the key dimension. Multi-head attention applies H parallel attention heads and concatenates the results:

$$\text{MultiHead}(Q, K, V) = [h_1; h_2; \ldots; h_H]W^O, \quad h_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{1.6}$$

This formulation allows each token to attend directly to any other token in the sequence regardless of distance, eliminating the vanishing gradient problem that limited recurrent networks.

BERT [4] established the pre-train-then-fine-tune paradigm for NLP. The pre-training phase uses two self-supervised objectives on large unlabelled corpora. The Masked Language Modelling (MLM) objective randomly masks 15 % of input tokens and trains the model to predict the original tokens from context:

$$\mathcal{L}_{\text{MLM}} = -\sum_{i \in \mathcal{M}} \log P(x_i \mid \mathbf{x}_{\setminus \mathcal{M}}) \tag{1.7}$$

where $\mathcal{M}$ is the set of masked positions. The Next Sentence Prediction (NSP) objective trains the model to classify whether two sentence segments are consecutive in the source document. After pre-training, a lightweight task-specific head (a single linear layer for token classification) is attached, and the entire model is fine-tuned on the labelled task data. BERT-base achieves F1 = 0.913 on CoNLL-2003, substantially surpassing BiLSTM-CRF baselines.

RoBERTa improved upon BERT by removing the NSP objective (shown to be harmful in subsequent studies), training with larger batch sizes (8 K vs. 256), longer training (160 GB of text vs. 16 GB), and dynamic masking that regenerates the mask pattern for each epoch. These changes produce more stable and higher-quality representations: RoBERTa-large achieves F1 = 0.93 on CoNLL-2003 [5].

XLM-RoBERTa (XLM-R) [6] extends the RoBERTa pre-training procedure to 100 languages by training on 2.5 TB of filtered CommonCrawl data. The shared sub-word vocabulary of 250 000 SentencePiece tokens enables cross-lingual knowledge transfer: representations of semantically equivalent expressions in different languages cluster together

in the embedding space. XLM-R base contains 12 transformer layers, 768 hidden dimensions, 12 attention heads, and approximately 125 million parameters. Its coverage of both Kazakh and Russian within a single unified vocabulary makes it particularly well-suited for the code-switching characteristics of the CTS corpus. A formal study by Conneau et al. [6] demonstrates that XLM-R substantially outperforms mBERT on NER tasks across 16 languages, including morphologically rich Slavic and Turkic languages.

For the NER task, the XLM-R encoder is extended with a token classification head: a linear layer $W \in \mathbb{R}^{768 \times K}$ that projects each contextual token embedding $\mathbf{h}_i \in \mathbb{R}^{768}$ to a vector of class logits, followed by a softmax to obtain label probabilities:

$$p(y_i = c \mid \mathbf{x}) = \frac{\exp(\mathbf{h}_i^\top \mathbf{w}_c)}{\sum_{j=1}^{K} \exp(\mathbf{h}_i^\top \mathbf{w}_j)} \tag{1.8}$$

where $K = 13$ is the number of entity labels (6 entity types $\times$ 2 BIO prefixes $+$ O).

1.3.3 Embedding-based pipelines (E5, multilingual models)

A parallel line of research develops dense text embeddings optimised for semantic similarity and retrieval through contrastive or instruction-tuned training, rather than masked language modelling. These models produce representations in which semantically similar texts are geometrically proximate in the embedding space, independently of surface form.

Multilingual-E5 (mE5) [7] is a family of XLM-RoBERTa-based models trained on approximately one billion multilingual text pairs using a two-stage procedure: weak supervision from noisy parallel data, followed by fine-tuning on high-quality supervised pairs from MS MARCO, Natural Questions, and TriviaQA. The training objective is a contrastive loss that pulls together representations of semantically equivalent text pairs and pushes apart representations of non-matching pairs:

$$\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau)}{\exp(\text{sim}(\mathbf{q}, \mathbf{d}^+)/\tau) + \sum_j \exp(\text{sim}(\mathbf{q}, \mathbf{d}_j^-)/\tau)} \tag{1.9}$$

where $\mathbf{q}$ is a query embedding, $\mathbf{d}^+$ is a positive document, $\mathbf{d}_j^-$ are in-batch negatives, and τ is a temperature parameter.

Although primarily designed for sentence-level retrieval, mE5's contextual token representations can be adapted for token-level NER by replacing the sentence-level pooling with token-level projection. The `multilingual-e5-base` variant (278M parameters) was selected for this thesis as a middle ground between representation quality and computational cost.

Petasis and Karkaletsis [8] demonstrate that E5-large embeddings combined with an ensemble of classical classifiers (XGBoost, k-NN, Logistic Regression) via soft voting achieves accuracy $= 0.92$ and F1 $= 0.91$ on a low-resource Greek NER task, substantially outperforming fine-tuned transformer baselines. This result suggests that strong pre-trained embeddings can compensate for limited task-specific labelled data, which is directly relevant to the rare entity types in the CTS corpus.

1.3.4 Large language models for few-shot NER

Recent years have seen the emergence of instruction-tuned large language models (LLMs) — models with hundreds of billions of parameters trained on vast multi-task corpora, enabling flexible task adaptation through natural language prompting rather than gradient-based fine-tuning. The few-shot paradigm [9] provides the model with a small number of task-specific examples in the prompt and asks it to generalise to new inputs without any weight updates.

In-context learning for NER frames the extraction task as a conditional text generation problem: given the system prompt, few-shot examples, and the input complaint text, the model generates a JSON object mapping entity types to extracted spans. This approach requires no labelled training data beyond the few-shot examples, making it attractive for entity types with very few training instances.

Qwen2.5 [10] is an open-weight LLM series (0.5B–72B parameters) developed by Alibaba. The 7B-parameter instruction-tuned variant supports 29+ languages including Russian and Kazakh, and demonstrates strong structured output generation through instruction tuning on supervised fine-tuning (SFT) and Reinforcement Learning from Human Feedback (RLHF) datasets. Instruction tuning is critical for JSON-structured NER outputs: a model must not only identify entity spans but also reliably format the output as valid JSON with correct field names.

Comparative LLM performance on NER. Keraghel et al. [5] report that GPT-4 in zero-shot mode achieves F1 $= 0.83$ on CoNLL-2003, narrowing to within 10 percentage points of fine-tuned RoBERTa-large (F1 $= 0.93$). LLaMA3.1 and Mistral achieve lower performance (F1 ≈ 0.60–0.70) on the same benchmark, particularly on entity types with limited lexical cues. In the present thesis, four LLMs are evaluated under identical few-shot prompting conditions to isolate the effect of model capacity and multilingual pre-training coverage.

Prompt construction critically influences LLM-based NER performance. The system prompt defines the entity taxonomy and provides extraction rules in natural lan-

guage. In-context examples are selected to ensure coverage of all entity types, with particular attention to the rarest classes (`STREET_INTERSECTION`, `STREET_NAME`) that the model is unlikely to have seen in sufficient context during pre-training. Post-processing filters remove hallucinated spans — a common failure mode in which the model generates plausible-looking entity values that do not appear in the source text.

1.4 Using machine learning and natural language processing for multilingual text analysis

Multilingual NLP addresses the challenge of building language technology for multiple languages simultaneously, particularly for language pairs that are underrepresented in standard benchmarks. The Kazakh-Russian bilingual setting presents several specific challenges that must be understood and addressed in the system design.

Code-switching is the intra-sentence mixing of two or more languages within a single utterance. Code-switching is pervasive in the CTS corpus: approximately 67 % of complaint records contain at least one sentence-internal language switch. A representative example is: «м/а 48, уақыты 09:30, остановка Тауелсиздик бағытында жоқ» (route 48, time 09:30, no stop in the direction of Tauelsizidik), which mixes Russian transit terminology with Kazakh temporal and directional phrases. Code-switching violates the monolingual assumptions of models pre-trained on single-language corpora and can cause out-of-vocabulary failures or degraded representations for tokens of the less-covered language.

Kazakh morphological complexity. Kazakh is an agglutinative language in which grammatical information (case, number, possession, tense) is encoded through a series of suffixes appended to the root form. A single noun may appear in up to 9 grammatical cases, each producing a distinct surface form. For geotag recognition, this means that a stop name such as «Байтурсынов» may appear in context as «Байтурсыновта» (locative: at Baitursunov), «Байтурсыновқа» (dative: towards Baitursunov), or «Байтурсыновтан» (ablative: from Baitursunov). Sub-word tokenisation (SentencePiece BPE) partially addresses this by segmenting inflected forms into root and suffix subword pieces, but the assignment of BIO labels must be done at the word level rather than the subword level to maintain annotation consistency.

Russian morphological complexity. Russian is a fusional language with a six-case declension system and grammatical gender that affects adjective agreement. Street names in Russian appear in various case forms depending on their syntactic role: «про-

спект Абая» (nominative) versus «на проспекте Абая» (prepositional) versus «по проспекту Абая» (dative). The presence of a case-inflected preposition does not change the entity boundary, which must always include the full canonical form of the street name.

Low-resource asymmetry. Kazakh is significantly lower-resourced than Russian in most multilingual pre-training corpora. While Russian is one of the top ten most represented languages in CommonCrawl, Kazakh representation is considerably smaller. This asymmetry means that multilingual models may allocate insufficient model capacity to Kazakh morphological patterns, leading to suboptimal sub-word representations for Kazakh tokens in mixed-language input.

Cross-lingual transfer learning is the mechanism by which multilingual models apply knowledge learned from one language to improve performance on another. XLM-RoBERTa's shared sub-word vocabulary creates an implicit alignment between representations of semantically related tokens across languages: the representation of «маршрут» (Russian: route) is similar to «бағыт» (Kazakh: route), even if the surface forms are entirely different. This cross-lingual alignment is crucial for the CTS task, where the same entity type can be expressed in either language.

Iqbal et al. [11] demonstrate that multilingual BERT (mBERT) achieves macro F1 = 0.98 on an augmented multilingual NER dataset, outperforming monolingual RoBERTa (F1 = 0.884). The authors attribute this advantage to cross-lingual transfer rather than simply greater model capacity, confirming that multilingual pre-training is beneficial even for individual-language NER tasks. Kassab et al. [12] show that DeepPavlov RuBERT remains the strongest single Russian-language NER model, achieving 28 % precision improvement over commercial alternatives through its domain-specific pre-training on Russian internet text. However, RuBERT's monolingual scope makes it fundamentally unsuited for the Kazakh-Russian mixed corpus without significant additional adaptation.

Tao et al. [13] demonstrate that BERT + BiLSTM + CRF with NLP-specific preprocessing achieves F1 = 0.93 on geographically annotated social media data in Chinese, demonstrating that contextual embeddings combined with preprocessing for informal register substantially boost accuracy on noisy, user-generated content. This finding is directly relevant to the CTS corpus, which shares many characteristics with social media text: informal register, abbreviations, inconsistent punctuation, and entity references embedded in unstructured prose.

Geoparsing — the combined task of geographic entity recognition and coordinate resolution — extends NER with a disambiguation step that links extracted toponyms to

specific coordinates in a geographic knowledge base. Hu et al. [14] evaluate 27 geoparsing tools across 26 public datasets and find that BERT-based models achieve the highest F1 on social media text (TopoBERT F1 = 0.854) while classical tools such as Stanford NER degrade sharply on noisy input. Moncla et al. [15] address toponym ambiguity resolution using CamemBERT and XLM-RoBERTa on diverse French geographic corpora, reporting F1 ≈ 0.92, and establish that neural classifiers on transformer embeddings effectively resolve spatial entity ambiguity that causes rule-based systems to fail.

Sun et al. [16] demonstrate that incremental domain-specific pre-training of BERT on Chinese social media text, combined with CRF and error correction post-processing (ChineseCTRE), improves geographic NER recall by 4–5 percentage points over base BERT, confirming that domain adaptation of pre-trained models is a viable strategy for improving minority-class performance in domain-specific corpora.

1.5 Comparison of NER architectures for low-resource and code-switched (Russian–Kazakh) settings

Table 1.2 summarises the 21 key related works surveyed for this thesis, spanning rule-based, statistical, encoder, embedding, and LLM-based NER approaches. The works are presented in the order introduced in the thesis presentation.

Table 1.2: Summary of key related works on NER and geotag extraction.

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
1	Yang et al. (2023) [9]	BERT, GPT-2/3/4, T5, LLaMA, BLOOM	Survey (no dataset)	Covers evolution from BERT to ChatGPT across pretraining objectives and NLP tasks
2	Keraghel et al. (2024) [5]	BiLSTM-CRF, BERT, RoBERTa, GPT-3/4, LLaMA	RoBERTa F1 = 0.93; GPT-4 zero-shot = 0.83	Fine-tuned encoders dominate supervised NER; LLMs excel in few-shot settings
3	Hu et al. (2023) [14]	27 tools incl. TopoBERT, spaCy, Stanford NER	TopoBERT F1 = 0.854	BERT-based models best on social media; classical tools degrade sharply on noisy text
4	Iqbal et al. (2024) [11]	mBERT, RoBERTa-small/large, BERT-base/large	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on augmented multilingual NER

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Table 1.2 continued from previous page

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
5	Kassab et al. (2025) [12]	DeepPavlov RuBERT, Natasha, Stanza, spaCy	+28% precision over single RuBERT	Semi-automated multi-model voting reduces manual annotation effort drastically
6	Petasis & Karkaletsis (2024) [8]	E5-large + XGBoost, KNN, LogReg (soft voting)	Accuracy = 0.92; F1 = 0.91	E5 embeddings + ML ensemble beats fine-tuned transformers on low-resource NER
7	Wang et al. (2023) [7]	mE5-small/base/large/large-instruct	BEIR score = 64.4 (large-instruct)	Two-stage contrastive training yields strong multilingual retrieval embeddings
8	Yang et al. (2024) [10]	Qwen2.5 (0.5B–72B, Base, Instruct, MoE)	MMLU, HumanEval benchmarks	Instruction tuning improves structured JSON output; open-weight models cost-effective
9	Mai et al. (2024) [17]	Mistral, Baichuan2, Llama 2, Falcon + geo-KB	Improved geoparsing precision/fairness	LLM fused with geographic knowledge bases improves geoparsing for underrepresented regions
10	Tarannum et al. (2026) [18]	mBERT, RoBERTa-small, BERT-large/base	mBERT macro F1 = 0.98	Multilingual BERT outperforms monolingual RoBERTa on heterogeneous address data
11	Wei et al. (2024) [19]	ALBERT-Conv1D-BiLSTM-CRF	P = 0.981; R = 0.967; F1 = 0.974	Fusing global (ALBERT) and local (Conv1D) features improves historical toponym NER
12	Moncla et al. (2022) [15]	CamemBERT, mBERT, XLM-RoBERTa + classifier	CamemBERT F1 ≈ 0.92	Neural classifiers on contextual transformer embeddings resolve toponym ambiguity
13	Tao et al. (2022) [13]	BERT + BiLSTM + CRF + NLP preprocessing	F1 = 0.93	Enhanced BERT with preprocessing excels on noisy social media geographic NER
14	Manjavacas et al. (2026) [20]	GPT-4o, GPT-OSS-120B, Llama-3.1-8B	GPT-4o ≈100%; Llama ≈73%	Larger models handle complex narrative NER fields better than smaller open-weight models
15	Stankevičius & Lukoševičius (2024) [21]	BERT-110M, T5, Random Embeddings	Repr.-shaping ≈0.70 vs. baseline ≈0.50	Simple representation shaping can match complex BERT models on similarity benchmarks
16	Mansourian & Oucheikh (2024) [22]	Llama 2 (LangChain + RAG) + NLP extraction	ChatGeoAI ≈25% on complex tasks	LLMs need RAG and entity mapping to execute complex geospatial workflow tasks

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Table 1.2 continued from previous page

No.	Author(s) & Year	Model(s) Used	Key Metric	Key Finding
17	Lutsai & Lampert (2024) [23]	Multilingual BERT + multitask PMOP regression	PMOP: 96% acc.; 3.68 km median error	Custom multitask loss with PMOP achieves lowest spatial errors in tweet geolocation
18	Alsaqer et al. (2023) [24]	LR, SVM, RF, Naïve Bayes	Accuracy = 67%	Classical ML predicts tweet location from text; accuracy limited by short noisy text
19	Santos et al. (2021) [25]	BERT-based vs. HMM/CRF baselines	BERT F1 ≈ 92–95%; HMM/CRF ≈ 80–85%	BERT significantly outperforms HMM/CRF for parsing unstructured real-world address text
20	Sun et al. (2023) [16]	ChineseCTRE (BERT + BiLSTM-CRF, incr. pre-training)	F1 ≈ 91–92% vs. baseline 87–88%	Incremental domain pre-training with error correction boosts geographic NER recall
21	Ussenbayeva et al. (2026) [1]	XLM-RoBERTa (base), Regex, CRF, Frozen LM	ROUTE_NUM F1 = 0.920; STREET_NAME F1 = 0.282	Transport NER is heterogeneous; aggregated metrics conceal per-class disparities; composite entities require specialised modelling

Synthesis of literature findings. The literature review reveals several consistent patterns across all surveyed works that directly inform the architectural choices made in this thesis.

Transformer encoders dominate supervised NER. Across all evaluation settings with sufficient labelled data, fine-tuned transformer encoders (BERT, RoBERTa, XLM-RoBERTa) consistently outperform both rule-based systems and classical statistical models (CRF, BiLSTM-CRF). The performance gap is particularly pronounced on noisy, informal text (social media, complaint records) where handcrafted features and gazetteers fail to generalise.

Multilinguality is non-negotiable for code-switching corpora. Works 4, 10, and 12 collectively establish that multilingual models (mBERT, XLM-RoBERTa) substantially outperform monolingual models when the evaluation corpus contains mixed-language content, even when the monolingual model is specifically tuned for one of the target languages. This finding has direct implications for the RuBERT baseline in this thesis, which is expected to underperform on Kazakh-dominant sentences.

LLMs are competitive on rare entity types. Works 1, 2, 8, 9, and 14 show that instruction-tuned LLMs are particularly well-suited to entity types with few training examples, where fine-tuned encoders suffer from overfitting or label collapse. In the CTS

corpus, `STREET_INTERSECTION` has only 2 training instances, making LLM few-shot extraction the only viable approach for this entity type.

Aggregate metrics conceal category-level heterogeneity. Work 21 (the authors' own prior publication) explicitly demonstrates that a single micro-averaged F1 score can give a misleadingly optimistic impression of system performance on a heterogeneous entity taxonomy. Per-class evaluation is essential for production deployment, where the operational cost of misclassifying a `STOP_NAME` differs substantially from missing a `STREET_INTERSECTION`.

Gap analysis. No prior work targets the specific combination of (a) Kazakh-Russian code-switching, (b) transport-domain operational complaint text, (c) a six-class entity taxonomy including composite spatial entities, and (d) a comparative evaluation of four architecturally distinct approaches under identical experimental conditions. This gap defines the contribution of the present thesis.

1.6 Setting the task

1.6.1 Formulation of the purpose and objectives of the thesis on automatic geotag recognition

Research Aim: To develop and evaluate multiple machine learning approaches for automatic geotag recognition in multilingual (Kazakh-Russian) textual data and to compare their effectiveness in terms of accuracy and robustness across structurally heterogeneous entity types.

Research Object: The process of named entity recognition in multilingual public transport complaint texts collected by City Transportation Systems (CTS) of Astana, Kazakhstan.

Research Subject: Machine learning models and algorithms for automatic extraction of six domain-specific geotag entity types from noisy, code-switched Kazakh-Russian natural language texts.

Objectives:

1. Conduct a systematic analysis of current methods for automatic geotag recognition and named entity recognition, identifying the state of the art in multilingual and low-resource settings.

2. Collect, preprocess, and annotate a domain-specific multilingual dataset of public transport complaint records with gold-standard BIO labels for six entity types.
3. Design and implement four machine learning approaches for geotag recognition: fine-tuned XLM-RoBERTa, fine-tuned RuBERT, fine-tuned Multilingual-E5, and a few-shot LLM pipeline.
4. Implement and evaluate four baseline methods (regex, frozen XLM-R linear probe, CRF, most-frequent class) to contextualise model performance.
5. Evaluate and compare all models using per-class precision, recall, F1-score, confusion matrix analysis, and boundary stability analysis.
6. Identify the most effective approach for each entity type and the overall best strategy for deployment in a real-time operational environment.

1.6.2 Requirements for an intelligent geotag-extraction system

The intended deployment context — real-time triage of CTS complaint records — imposes the following functional and non-functional requirements on the geotag-extraction system:

1. **Multilingual support.** The system must process Kazakh, Russian, and code-switched sentences without explicit language identification as a prerequisite step.
2. **Six-class entity recognition.** The system must classify extracted spans into the six domain-specific entity types: `ROUTE_NUM`, `BUS_ID`, `PLATE_NUM`, `STOP_NAME`, `STREET_NAME`, `STREET_INTERSECTION`.
3. **Span-level precision ≥ 0.70.** Downstream geocoding workflows are sensitive to false positives; high precision is the primary operational constraint to prevent spurious alert generation.
4. **Recall coverage of `STOP_NAME` and `ROUTE_NUM` ≥ 0.80.** These are the most frequent and operationally critical entities, forming the primary inputs to dispatch routing and hotspot detection.
5. **Computational throughput.** For real-time deployment on standard server hardware, the system should be capable of processing at least 100 records per minute, setting a practical limit on model size and inference latency.

6. **Graceful degradation on noisy input.** The system must tolerate spelling errors, missing punctuation, non-standard abbreviations, and mixed-script input that are characteristic of citizen complaint text.

7. **Structured output.** Extracted entities must be returned as structured data (entity type, extracted text span, character offsets) suitable for downstream GIS linking and database insertion.

1.6.3 Formulation of data requirements for model training

Formal task definition. We formulate geotag recognition as a token-level sequence labelling task under the BIO (Beginning-Inside-Outside) tagging scheme. Given an input token sequence $\mathbf{T} = (t_1, t_2, \ldots, t_n)$, the objective is to assign each token t_i a label $l_i \in \mathcal{L}$, where:

$$\mathcal{L} = \{B\text{-}e,\ I\text{-}e \mid e \in \mathcal{E}\} \cup \{O\}, \quad |\mathcal{L}| = 2|\mathcal{E}| + 1 = 13 \tag{1.10}$$

$\mathcal{E} = \{\texttt{ROUTE_NUM}, \texttt{BUS_ID}, \texttt{PLATE_NUM}, \texttt{STOP_NAME}, \texttt{STREET_NAME}, \texttt{STREET_INTERSECTION}\}$ denotes the set of six entity types, and O denotes the outside class. The BIO scheme encodes entity spans such that the first token of an entity receives the B- prefix (beginning), all subsequent tokens within the same span receive the I- prefix (inside), and all non-entity tokens receive O (outside). A predicted entity span $\hat{s}$ is considered a true positive only if both its left and right boundaries exactly match the gold annotation and its entity type label is correct.

BIO annotation constraints. The BIO scheme enforces the following structural constraints on valid label sequences:

- An I-e label must be immediately preceded by either B-e or I-e (for the same entity type e).
- B-e may follow any label (including I-e' for a different entity type $e' \neq e$, which terminates e' and begins e).
- O terminates any open entity span.

Data requirements.

- A minimum of 2 000 labelled complaint records to support fine-tuning of transformer encoders without severe overfitting.

- Stratified splitting to preserve entity type proportions across training and test sets, ensuring that the test distribution is representative of the full corpus.
- Gold-standard annotation produced by at least two independent annotators with native or near-native proficiency in both Kazakh and Russian, with adjudication of disagreements.
- Minimum coverage of the rarest entity type: at least 50 training instances of `STREET_INTERSECTION` to enable meaningful fine-tuning of minority-class behaviour.
- The annotated data must be stored in JSONL format with pre-tokenised BIO label sequences to support efficient DataLoader-based training without on-the-fly tokenisation.

Class imbalance handling. Given the severe skew of the entity distribution (Section 1.2), the following strategies are required to prevent label collapse on minority classes:

- *Weighted cross-entropy loss*: minority entity classes receive higher loss weights, computed as $w_c = N_{\text{total}}/(K \cdot N_c)$ where N_c is the count of class c.
- *Oversampling*: training records containing rare entity types are duplicated k times in the training pool, increasing their effective representation without collecting new data.
- *Macro-averaged evaluation*: the primary evaluation metric weights all entity types equally, ensuring that strong performance on frequent classes does not mask failure on rare ones.

1.6.4 Defining criteria for evaluating the effectiveness of NER models

Model performance is assessed using standard span-level NER metrics computed by the `seqeval` library, which requires *exact boundary and label match* for a true positive. A predicted span $\hat{s} = (\hat{t}, \hat{l}, \hat{r})$ (type, left boundary, right boundary) is a true positive if and only if there exists a gold span $s^* = (t^*, l^*, r^*)$ such that $\hat{t} = t^*$, $\hat{l} = l^*$, and $\hat{r} = r^*$ simultaneously.

Per-class precision, recall, and F1:

$$\mathrm{P}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \quad \mathrm{R}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \quad F_{1,c} = 2 \cdot \frac{\mathrm{P}_c \cdot \mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c} \tag{1.11}$$

Macro-averaged F1 weights all entity types equally regardless of frequency:

$$F_{1,\mathrm{macro}} = \frac{1}{|\mathcal{E}|} \sum_{c \in \mathcal{E}} F_{1,c} \tag{1.12}$$

Macro-averaged F1 is the primary evaluation metric because it is sensitive to performance on rare classes such as `STREET_INTERSECTION` and `STREET_NAME`.

Micro-averaged F1 aggregates counts across all classes before computing the ratio:

$$F_{1,\mathrm{micro}} = 2 \cdot \frac{\sum_c \mathrm{TP}_c}{\sum_c (2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c)} \tag{1.13}$$

Micro-averaged F1 is influenced primarily by frequent classes and provides an overall measure of throughput-weighted performance.

Boundary stability measures the fraction of correctly labelled predicted entity spans that also have correct boundaries:

$$\mathrm{BS}_c = \frac{|\{(t,l,r) \in \hat{S}_c : \exists\,(t^*,l^*,r^*) \in S_c^*,\ l = l^* \wedge r = r^*\}|}{|\hat{S}_c \cap S_c^*|_{\text{type-only}}} \tag{1.14}$$

where $|\hat{S}_c \cap S_c^*|_{\text{type-only}}$ counts spans that match in type but not necessarily in boundary. High boundary stability with high precision indicates a model that correctly identifies *what* an entity is but may over- or under-extend its span. Low boundary stability indicates fundamental span segmentation errors that would propagate errors to downstream geocoding.

Confusion matrix analysis at the entity-type level (collapsing BIO prefixes) identifies systematic misclassification patterns. For the CTS entity taxonomy, the most informative confusion axis is between `BUS_ID` and `PLATE_NUM` (both short alphanumeric identifiers) and between `STOP_NAME` and `STREET_NAME` (both multi-word spatial toponyms). Confusion between these pairs indicates that the model has learned surface form rather than functional context.

Together, these criteria provide a comprehensive and operationally motivated evaluation framework that goes beyond a single aggregate score and enables category-aware model selection for deployment.

Chapter 2

Model development and experimental setup

This chapter describes the complete experimental pipeline developed for the geotag recognition task. Section 2.1 documents the data collection, manual annotation, preprocessing, and dataset partitioning that produce the gold-standard corpus used throughout this thesis. Section 2.2 presents the four machine learning approaches under evaluation, motivating each architectural choice, specifying the training objectives, and detailing the hyperparameter configurations. Section 2.3 defines four reference baselines of increasing sophistication that contextualise the main results. Section 2.4 formalises the evaluation protocol and explains why each metric was chosen. Together, these sections provide a fully reproducible specification of the experimental setup, enabling direct comparison with future work on the same or related corpora.

2.1 Data collection and preparation

2.1.1 Data collection: transport complaint records from City Transportation Systems

The primary data source is the operational complaint database of City Transportation Systems (CTS), the public transport operator of Astana, Kazakhstan. CTS processes approximately 3 000 complaint records per month through three intake channels: a citizen-facing mobile application, a municipal web portal, and a call-centre transcription service. Each record contains a mandatory free-text description field (*описание*) in which passengers describe transportation incidents including missed stops, route deviations, driver misconduct, payment failures, and vehicle condition reports. The records were collected

over a twelve-month operational period and anonymised by CTS prior to transfer, with all personally identifiable information removed. No geolocation metadata was included alongside the records, making the free-text description field the sole source of spatial information for every complaint.

The raw corpus presents several characteristics that distinguish it sharply from standard NLP benchmarks such as CoNLL-2003 or OntoNotes. First, complaint texts are written predominantly in Russian and Kazakh, with extensive intra-sentence code-switching: approximately 67 % of records contain at least one sentence-internal language switch, and in many records the two languages alternate within a single clause. A representative example is: «маршрут 48, уақыты 09:30, остановка Тауелсиздік бағытында жоқ» (route 48, time 09:30, stop in the direction of Tauelsizidik absent), which combines Russian transit vocabulary with Kazakh temporal and directional phrases. Second, the register is informal and spontaneous, exhibiting abbreviations («м/а» for «маршрут/автобус», «ул.» for «улица»), non-standard punctuation, phonetic spelling, and mixed-script alphanumeric vehicle identifiers in which Cyrillic and Latin characters are interleaved. Third, the entity distribution is severely skewed: route numbers and stop names collectively account for approximately 65 % of all entity occurrences, while street intersections represent under 1 % of all annotated spans. These properties collectively define a low-resource, noisy, multilingual NER scenario that departs substantially from the well-studied benchmark settings in which the dominant model architectures were originally developed and evaluated.

2.1.2 Manual BIO annotation in Label Studio

Manual annotation was performed using *Label Studio*, an open-source data-labelling platform that supports token-level span annotation with configurable label schemas. Three annotators with native or near-native proficiency in both Kazakh and Russian independently labelled each complaint record according to the six-class entity taxonomy defined in Section 1.2 of Chapter 1. Prior to the main annotation campaign, a detailed annotation guideline was developed iteratively over a two-week pilot phase, specifying span boundary conventions, label priority rules for structurally ambiguous overlapping spans (e.g., a stop name that contains a street name as a substring), and canonical treatment of abbreviations and morphologically inflected forms. The guideline was revised after each of three pilot rounds until inter-annotator agreement on pilot batches of 50 records exceeded $\kappa = 0.80$.

Tokens were labelled under the BIO (Beginning-Inside-Outside) scheme. The first

token of each entity span received a B-prefix tag; all subsequent tokens within the same span received an I-prefix tag of the same entity type; all non-entity tokens received the O label. Annotators applied labels directly on the whitespace-tokenised text, selecting exact character-level substrings through the Label Studio span-selection interface. Annotation quality was controlled through an inter-annotator agreement (IAA) protocol applied after each batch of 200 records. Pairwise Cohen's κ [?] was computed at the span level for each entity type, requiring simultaneous match on entity type, left boundary, and right boundary. The overall observed agreement reached $\kappa_{\text{overall}} = 0.84$, which falls in the "strong agreement" range ($\kappa > 0.80$) established by Landis and Koch [?]. Per-class values ranged from $\kappa = 0.91$ for `ROUTE_NUM` — whose high structural regularity (short numeric spans) leaves little room for boundary ambiguity — to $\kappa = 0.72$ for `STREET_INTERSECTION`, which spans 5–12 tokens and involves subjective decisions about whether directional qualifiers should be included within the entity boundary.

Disagreements were resolved using a majority-vote protocol: any span on which at least two of the three annotators agreed was accepted as gold standard without modification. Spans on which all three annotators produced different labels were escalated to the project supervisor for adjudication. This three-way disagreement scenario affected fewer than 2 % of all entity annotations, demonstrating that the guideline successfully eliminated the majority of systematic labelling ambiguities.

The final annotated dataset comprises 2 750 complaint records exported in JSONL format, with each record storing the raw text, the whitespace-tokenised word sequence, and the corresponding BIO label sequence. Table 2.1 summarises the annotation effort, corpus statistics, and agreement figures.

Table 2.1: Annotation statistics for the CTS geotag corpus.

Property	**Value**
Total complaint records annotated	2 750
Total annotated tokens (all labels)	73 821
Entity token proportion	14.8 %
Outside (`O`) token proportion	85.2 %
Number of annotators	3
Overall Cohen's κ	0.84
Three-way disagreement rate (adjudicated)	< 2 %
Annotation tool	Label Studio
Export format	JSONL (BIO)

The statistics in Table 2.1 highlight two characteristics of the corpus that have direct consequences for model design. The entity token proportion of only 14.8 % means that 85.2 % of all tokens carry the `O` label, creating a class imbalance of nearly 6:1 at the token level even before the disparity *among* entity classes is considered. A model that assigns `O` to every token would achieve overall token-level accuracy of 85.2 % while extracting no entities whatsoever, illustrating why token-level accuracy is a wholly inappropriate evaluation metric for this task. The overall $\kappa = 0.84$ confirms that the annotation is of high quality and that the gold-standard labels reflect genuine linguistic consensus rather than artefacts of individual annotator preference; this in turn justifies using a single adjudicated label sequence as the evaluation reference rather than measuring model performance against multiple competing annotations.

2.1.3 Preprocessing pipeline

Raw complaint text was transformed into model-ready input through a five-stage preprocessing pipeline. Each stage was applied uniformly across all four models prior to tokenisation, ensuring that any observed differences in model performance are attributable to architectural and training choices rather than to input-level treatment differences.

1. **Unicode normalisation.** Cyrillic characters were normalised to the canonical NFC Unicode form using Python's `unicodedata.normalize` function. Look-alike Latin-Cyrillic homoglyphs introduced by mixed-keyboard mobile typing (e.g.,

Latin ‘c’ (U+0063) used in place of Cyrillic ‘с’ (U+0441)) were detected using Unicode codepoint-range checks and silently replaced with the corresponding Cyrillic forms. This step is operationally critical because homoglyph substitution is widespread in citizen complaint text produced on mobile devices, and undetected homoglyphs produce out-of-vocabulary sub-words that degrade contextual representations.

2. **Whitespace tokenisation.** Records were split on whitespace boundaries and Unicode punctuation characters. Punctuation tokens were retained as separate items to preserve complaint structure, since punctuation boundaries frequently coincide with entity span boundaries (e.g., a comma immediately following a stop name correctly delimits the entity’s right edge).

3. **Subword alignment.** Transformer models operate on sub-word tokens produced by model-specific sub-word tokenisers: SentencePiece BPE [**?**] for XLM-R and mE5 (vocabulary size 250 000), and WordPiece [**?**] for ruBERT. Word-level BIO labels must be projected onto sub-word tokens. For a word w_i tokenised into sub-words $s_1, s_2, \ldots, s_m$, the alignment rule assigns the original BIO label to s_1 and a special ignore value -100 to $s_2, \ldots, s_m$:

$$\ell'_k = \begin{cases} \ell_i & \text{if } s_k \text{ is the first sub-word of word } w_i \\ -100 & \text{otherwise} \end{cases} \tag{2.1}$$

The value -100 is excluded from the cross-entropy loss computation by the PyTorch `CrossEntropyLoss` implementation, ensuring that only the first sub-word of each word contributes to parameter updates. This convention follows the original BERT fine-tuning prescription [4] and prevents the model from receiving duplicate or contradictory gradient signals for multi-piece words.

4. **Sequence truncation.** Input sequences exceeding the model’s maximum context length were truncated to 512 sub-word tokens for XLM-R and mE5, and to 128 tokens for ruBERT. Corpus analysis showed that 98.4 % of CTS complaint records contain fewer than 128 whitespace-level tokens and 99.8 % contain fewer than 512 sub-word tokens, meaning that truncation affects at most 0.2 % of records for the primary models. The few truncated records contained repetitive complaint text in which the entity-bearing portion appeared in the first 512 tokens; no entity span boundaries crossed a truncation point in the analysis.

5. **Code-switching handling.** No explicit language identification or per-language text separation was performed. Multilingual models (XLM-R, mE5) process mixed-language tokens natively within a shared sub-word vocabulary, exploiting the cross-lingual alignment properties induced by multilingual pre-training. The LLM pipeline (Qwen2.5) handled code-switching at the prompt level through bilingual in-context examples that demonstrated entity extraction in both Kazakh-dominant and Russian-dominant sentences. This design choice deliberately avoids the error propagation associated with upstream language identification, which is itself an unsolved problem for heavily code-switched conversational text [6].

2.1.4 Stratified train–test split

The annotated corpus was partitioned into training and test sets using stratified sampling to maintain entity type prevalence across splits. Stratification was applied at the record level, using the set of entity types present in each record as the stratification key. Records containing no entities were treated as a separate stratum, labelled `NO_ENTITY`, to ensure that entity-free examples are adequately represented in both splits and do not cluster artificially in either partition. The split was performed with a fixed random seed of 42 to ensure reproducibility.

The resulting partition yields 1 925 training records (70 %) and 825 test records (30 %). Table 2.2 presents entity-level statistics for each split, reporting the number of records containing at least one instance of each entity type (presence counts) and the total token counts per BIO label class.

Table 2.2: Dataset statistics by entity type after stratified 70/30 split. *Presence count* is the number of records containing at least one instance of the entity type; *tokens* counts all B- and I-labelled tokens of that entity type.

Entity Type	**Train presence**	**Test presence**	**Train tokens**	**Test tokens**	**Avg. tok**
`ROUTE_NUM`	1 619	646	2 641	1 131	
`BUS_ID`	71	312	798	342	
`PLATE_NUM`	491	132	623	267	
`STOP_NAME`	1 616	462	3 901	1 672	
`STREET_NAME`	34	167	287	123	
`STREET_INTERSECTION`	2	6	52	22	
Total entity tokens	–	–	8 302	3 557	
Total records	1 925	825	–	–	

Table 2.2 reveals three structural properties of the dataset that have direct implications for model training and evaluation. First, the `STREET_INTERSECTION` class is represented by only 2 training records and 52 training tokens, providing essentially no signal for gradient-based fine-tuning. Any encoder model trained on this data will default to predicting `O` for potential intersection spans, since the probability of encountering a positive training example of this type in any given batch is negligible. This observation motivates the inclusion of the few-shot LLM approach (Section 2.2.4), which can leverage pre-training knowledge of spatial language without requiring labelled examples.

Second, the `STREET_NAME` class exhibits a severe train–test distributional asymmetry: only 34 training records contain a street name, compared to 167 test records. This imbalance, which arises because street-naming complaints are more seasonal or geographically concentrated, creates a situation in which the test set is drawn from a substantially broader distribution than the training set for this entity type. Models that rely on lexical memorisation of known street names will be at a particular disadvantage on the test set. This observation motivates the additional manual upweighting of `STREET_NAME` in the XLM-R training objective (see Section 2.2.1).

Third, `BUS_ID` shows the opposite train–test asymmetry: 71 training records versus 312 test records. This suggests that bus ID complaints are more prevalent during certain operational periods and that the twelve-month collection window captured a phase-shift in complaint types. Models must therefore generalise to bus identifier formats that may not be fully represented in the training distribution.

2.2 Machine learning models

Four machine learning approaches were implemented and evaluated, spanning the spectrum from fine-tuned multilingual encoders to generation-based few-shot prompting. The shared BIO label set $\mathcal{L}$ contains 13 entity labels (B- and I-prefix for each of the 6 entity types) plus the O class, yielding $K = 13$ output classes. All fine-tuning experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM) using the Hugging Face `transformers` library [**?**]. The four approaches are ordered by increasing architectural distance from classical supervised learning: XLM-R and ruBERT are standard encoder fine-tuning methods, mE5 adapts a contrastively trained retrieval model to a discriminative objective, and Qwen2.5 replaces gradient-based adaptation entirely with in-context learning.

2.2.1 Model 1: XLM-RoBERTa fine-tuned for token classification

Motivation and architecture

XLM-RoBERTa (`xlm-roberta-base`) [6] is a multilingual transformer encoder pre-trained on 2.5 TB of filtered CommonCrawl text spanning 100 languages, using the RoBERTa masked language modelling (MLM) objective without the next-sentence prediction auxiliary task. Its shared SentencePiece vocabulary of 250 000 sub-word types covers both Kazakh and Russian within a unified token space, enabling cross-lingual representation alignment: semantically equivalent expressions in different languages are mapped to geometrically proximate points in the embedding space. This property is directly beneficial for the code-switching characteristics of the CTS corpus, in which the same entity (e.g., a bus route) may be expressed using Russian or Kazakh vocabulary in different records. The base variant contains 12 transformer layers, 768 hidden dimensions, 12 attention heads, and approximately 125 million parameters, providing a strong representational capacity within the memory budget of the available GPU hardware.

For token classification, the pre-trained encoder is extended with a linear classification head. Given the contextual embedding $\mathbf{h}_i \in \mathbb{R}^{768}$ produced for token t_i by the final encoder layer, the head computes a label probability distribution:

$$p(y_i = c \mid \mathbf{x}) = \frac{\exp(\mathbf{w}_c^\top \mathbf{h}_i + b_c)}{\sum_{k=1}^{K} \exp(\mathbf{w}_k^\top \mathbf{h}_i + b_k)} \tag{2.2}$$

where $\mathbf{w}_c \in \mathbb{R}^{768}$ and $b_c \in \mathbb{R}$ are the learnable parameters of the linear head for class c, and $K = 13$. The full model is fine-tuned end-to-end: both the encoder backbone and the classification head receive gradient updates during training.

Class-weighted training objective

Given the severe class imbalance documented in Table 2.2, standard unweighted cross-entropy training will drive the model towards the O-dominant optimum, suppressing the gradient signal for rare entity classes whose token counts are orders of magnitude smaller than the outside class. To address this, a weighted cross-entropy loss is employed:

$$\mathcal{L}_{\text{XLM-R}} = -\frac{1}{N_{\text{valid}}} \sum_{i=1}^{N_{\text{valid}}} w_{y_i} \log p(y_i \mid \mathbf{x}) \tag{2.3}$$

where N_{valid} is the number of non-masked tokens (tokens with $\ell' \neq -100$ per Equation 2.1), and the per-class weight w_c is computed using the standard inverse-frequency formula:

$$w_c = \frac{N_{\text{total}}}{K \cdot N_c} \tag{2.4}$$

with N_{total} denoting the total number of labelled (non-masked) tokens and N_c the frequency of label c in the training set. Equation (2.4) ensures that the expected summed contribution of each class to the total loss is approximately equal across all K classes, regardless of frequency. In addition, `STREET_NAME` tokens receive an additional manual upscaling factor of 1.6, applied on top of the inverse-frequency weight, to compensate for the train–test distributional mismatch identified in Section 2.1.4: the 4.9× excess of `STREET_NAME` records in the test set over the training set requires the model to generalise beyond the narrow set of street names observed during training, which is encouraged by elevated loss weighting during training.

Learning rate schedule

Training employs a linear warm-up followed by linear decay, a schedule that stabilises fine-tuning of large pre-trained models by preventing destructive updates to the pre-trained representations in the first training steps [4]:

$$\eta(t) = \begin{cases} \eta_{\max} \cdot \dfrac{t}{T_{\text{warm}}} & t \leq T_{\text{warm}} \\ \eta_{\max} \cdot \dfrac{T - t}{T - T_{\text{warm}}} & t > T_{\text{warm}} \end{cases} \tag{2.5}$$

where $\eta_{\max} = 2 \times 10^{-5}$, T is the total number of training steps (number of batches $\times$ epochs), and $T_{\text{warm}} = 0.06\,T$ (6 % warm-up proportion, corresponding to approximately 86 warm-up steps for the 1 925 training records at batch size 8 over 10 epochs).

Training configuration

Table 2.3 summarises the complete hyperparameter configuration for XLM-R fine-tuning.

Table 2.3: XLM-RoBERTa fine-tuning configuration.

Hyperparameter	**Value**
Base model	`xlm-roberta-base`
Parameters	≈125M
Tokeniser	SentencePiece BPE (vocab 250 000)
Max sequence length	512 sub-word tokens
Training epochs	10
Batch size	8
Optimiser	AdamW
Learning rate $\eta_{\max}$	2×10^{-5}
Weight decay	0.01
Warm-up proportion	6 %
Loss function	Weighted cross-entropy
`STREET_NAME` upweight	×1.6 (additional)
Hardware	NVIDIA Tesla T4 (16 GB VRAM)
Random seed	42

The choice of 10 training epochs reflects a trade-off between underfitting on rare classes and overfitting on frequent classes. Preliminary experiments on a held-out validation set (10 % of the training records, not included in the reported evaluation) showed that macro-averaged F1 on rare classes continued to improve up to approximately epoch 8 before plateauing, while micro-averaged F1 on frequent classes peaked earlier at epoch 5. The final epoch count of 10 was selected to maximise macro-averaged performance on the validation set, consistent with the primary evaluation criterion established in Section 2.4.

2.2.2 Model 2: RuBERT (monolingual Russian BERT)

Motivation

DeepPavlov RuBERT (`DeepPavlov/rubert-base-cased`) [26] is a BERT-base [4] model pre-trained exclusively on Russian Wikipedia and a curated Russian news corpus, comprising approximately 180 million parameters. Its inclusion in this study serves two complementary purposes. First, it constitutes a domain-relevant monolingual baseline: because the majority of CTS complaint text is written in Russian, even a monolingual Russian model may achieve reasonable performance on the Russian-dominant records. Second, and more importantly, it directly measures the performance penalty incurred by ignoring the Kazakh portion of the corpus. Because RuBERT's WordPiece vocabulary was constructed entirely from Russian text, Kazakh tokens that are absent from the vocabulary will be fragmented into individual character pieces, producing degraded sub-word representations that carry no morphological or semantic information about the Kazakh token. The expected performance gap between RuBERT and XLM-R on records with substantial Kazakh content thus provides empirical evidence for the value of cross-lingual pre-training in code-switching settings.

Subword alignment

RuBERT uses the WordPiece tokeniser, which produces a different sub-word segmentation from the SentencePiece tokeniser used by XLM-R. Despite this difference, the alignment rule of Equation (2.1) applies identically. Alignment is performed using the offset mapping produced by the Hugging Face `tokenizers` library, which maps each sub-word token to the character-level span of the original word it represents:

$$\ell'_k = \begin{cases} \ell_i & \text{if offset}(s_k) = \text{offset(first sub-word of } w_i) \\ -100 & \text{otherwise} \end{cases} \tag{2.6}$$

For Kazakh words absent from the Russian vocabulary, WordPiece produces `[UNK]` tokens or single-character pieces; in both cases, the alignment assigns the gold label to the first piece and -100 to all subsequent pieces, consistent with the treatment of normal multi-piece words.

Prediction and training configuration

Token label assignment is performed by:

$$\hat{y}_i = \arg\max_{c \in \mathcal{L}} \ \mathrm{softmax}(W\mathbf{h}_i + \mathbf{b})_c \tag{2.7}$$

where $W \in \mathbb{R}^{K \times 768}$ and $\mathbf{b} \in \mathbb{R}^K$ are the parameters of the classification head, and $\mathbf{h}_i$ is the contextual embedding from the final RuBERT encoder layer.

RuBERT is fine-tuned for 3 epochs with batch size 8, learning rate 2×10^{-5}, maximum sequence length 128, and AdamW optimiser with unweighted cross-entropy loss. The deliberately constrained training configuration serves a methodological purpose: it isolates the effect of pre-training language coverage from the confounding effects of extended training and class-weighting that are employed for XLM-R. A fairer direct comparison between RuBERT and XLM-R would require matching all training hyperparameters, but the purpose of RuBERT in this study is not to present the best-possible monolingual performance but rather to establish a controlled reference point that quantifies the linguistic coverage gap.

2.2.3 Model 3: Multilingual E5-base with token classification head

Motivation and architecture

Multilingual-E5-base (`intfloat/multilingual-e5-base`) [7] is an XLM-RoBERTa-based model comprising approximately 278 million parameters, trained through a two-stage contrastive learning procedure on approximately one billion multilingual text pairs. The first stage uses weakly supervised contrastive training on noisy parallel web data to induce a multilingual embedding space; the second stage fine-tunes the resulting model on high-quality supervised pairs from MS MARCO, Natural Questions, and MIRACL to sharpen the cross-lingual alignment. The contrastive training objective maximises the cosine similarity between semantically equivalent sentence pairs while pushing apart non-matching pairs:

$$\mathcal{L}_{\mathrm{E5}} = -\log \frac{\exp(\cos(\mathbf{q}, \mathbf{d}^+)/\tau)}{\exp(\cos(\mathbf{q}, \mathbf{d}^+)/\tau) + \sum_{j=1}^{M} \exp\big(\cos(\mathbf{q}, \mathbf{d}_j^-)/\tau\big)} \tag{2.8}$$

where $\mathbf{q}$ is the query embedding, $\mathbf{d}^+$ is the positive document embedding, $\mathbf{d}_j^-$ are M in-batch negatives, $\cos(\cdot, \cdot)$ denotes cosine similarity, and τ is a learnable temperature

parameter. This objective differs fundamentally from the MLM pre-training of XLM-R: mE5 learns to cluster semantically similar sentences rather than to reconstruct masked tokens. The key question explored by including mE5 is whether the stronger sentence-level semantic alignment induced by contrastive training yields better token-level representations for NER, or whether the MLM objective remains more beneficial for fine-grained token-level tasks.

Although designed for sentence-level retrieval, mE5's contextual token representations can be directly adapted for token-level NER by replacing sentence pooling with a token-level linear projection head. The forward pass for token classification is:

$$Z = XW + \mathbf{b}, \quad X \in \mathbb{R}^{N\times 768},\ W \in \mathbb{R}^{768\times K},\ Z \in \mathbb{R}^{N\times K} \tag{2.9}$$

where X is the matrix of contextual token embeddings from the final encoder layer and Z contains the raw class logits for the N tokens in the sequence.

Class-weight computation

Class weights for mE5 are computed exclusively over entity-class tokens (excluding O), then symmetrically clipped to prevent extreme values for near-absent classes:

$$w_c = \text{clip}\left(\frac{\sum_{j\in\mathcal{E}} N_j}{|\mathcal{E}|\cdot N_c},\ w_{\min},\ w_{\max}\right), \quad w_{\min} = 0.05,\ w_{\max} = 8.0 \tag{2.10}$$

The clipping upper bound of 8.0 prevents the near-zero count of `STREET_INTERSECTION` training tokens (52 tokens, Table 2.2) from producing weights in excess of 100, which would destabilise gradient norms and cause training divergence. The lower bound of 0.05 ensures that the most frequent entity class (here, `STOP_NAME`) still receives a meaningful loss contribution rather than being down-weighted to near-zero.

Training configuration

mE5 is fine-tuned for 15 epochs with batch size 8, gradient accumulation over 2 steps (effective batch size 16), learning rate 2×10^{-5}, AdamW optimiser (weight decay 0.01, warm-up ratio 0.1). The longer training schedule relative to XLM-R (15 vs. 10 epochs) compensates for the misalignment between mE5's contrastive pre-training objective and the discriminative NER fine-tuning objective: more gradient steps are required to adapt the top encoder layers from a sentence-similarity regime to a token-level boundary-detection

regime. The gradient accumulation of 2 steps is employed to maintain a stable effective batch size within the GPU memory budget while processing longer sequences generated by mE5's larger parameter count.

2.2.4 Model 4: Qwen2.5-7B via few-shot prompting

Motivation

Fine-tuned encoder models are fundamentally constrained by the quantity and quality of labelled training data available for each entity class. As established in Section 2.1.4 and quantified in Table 2.2, the `STREET_INTERSECTION` class contains only 2 training records, and `STREET_NAME` contains only 34. For these classes, gradient-based fine-tuning on the available data is expected to either fail entirely or produce unreliable boundary predictions. Few-shot prompting of large language models (LLMs) offers a principled alternative: the model exploits knowledge about spatial language and syntactic structure acquired during large-scale multilingual pre-training, requiring only a small number of in-context demonstrations to operationalise this knowledge for the target entity types.

Qwen2.5 [10] is an open-weight LLM series (0.5B–72B parameters) developed by Alibaba Cloud. The 7B instruction-tuned variant (`qwen2.5:7b-instruct`) supports more than 29 languages including Russian and Kazakh, and produces reliable structured JSON output through its supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) training stages. Instruction following is critical for this application: the model must not only identify entity spans but also format its output as valid JSON with correct field names and array syntax, failing gracefully when no entity of a given type is present. The model was deployed locally via the *Ollama* serving framework, enabling inference without external API calls or data transfer to third-party servers.

Prompt construction

The inference prompt for a complaint record x_i is constructed by concatenating three components:

$$P_i = \text{SYS} \oplus E_1 \oplus E_2 \oplus \cdots \oplus E_{12} \oplus x_i \tag{2.11}$$

where $\oplus$ denotes string concatenation, SYS is the system instruction block, and $E_1, \ldots, E_{12}$ are 12 in-context examples selected from the training set (2 per entity type, balanced across all 6 classes). The system instruction defines the entity taxonomy in natural language, provides extraction rules for both Russian and Kazakh surface forms, and specifies the required JSON output schema:

```
{"ROUTE_NUM": [...], "BUS_ID": [...], "PLATE_NUM": [...],
 "STOP_NAME": [...], "STREET_NAME": [...],
 "STREET_INTERSECTION": [...]}
```

In-context examples were selected to maximise coverage of structurally diverse surface forms, with particular care given to the rarest classes. For `STREET_INTERSECTION`, both available training instances were included as in-context examples; for `STREET_NAME`, examples were drawn from records with the highest annotator confidence (those on which all three annotators agreed). This selection strategy is designed to provide the model with the clearest possible signal for the entity types where task-specific demonstrations are scarcest.

Post-filtering

LLM-based extractors are known to produce hallucinated spans — entity values that appear plausible given the entity type but are absent from the source text [5]. A lightweight post-filter Φ retains only extracted spans satisfying three independently necessary conditions:

$$\hat{\mathcal{E}}_i^{(c)} = \left\{ v \in \tilde{\mathcal{E}}_i^{(c)} \;\middle|\; |v| \geq 2 \;\wedge\; v \notin \mathcal{W}_{\text{stop}} \;\wedge\; \phi_c(x_i) \right\} \tag{2.12}$$

where $\tilde{\mathcal{E}}_i^{(c)}$ is the raw model-extracted set for entity class c, $|v|$ is the character length of span v, $\mathcal{W}_{\text{stop}}$ is a manually curated set of noise tokens (function words, stopwords, and common abbreviations that cannot constitute valid entity values), and $\phi_c(x_i)$ is a class-specific keyword consistency check. For `STREET_INTERSECTION` specifically, a predicted span is retained only if the source text contains at least one spatial conjunction keyword («угол», «пересечение», «қиылысы»), since a genuine intersection reference requires this structural element. The character-length threshold ($|v| \geq 2$) eliminates single-character outputs that arise from JSON generation errors rather than genuine entity extraction. The post-filter is deliberately conservative: false negatives introduced by the filter are preferable to false positives, given that the downstream geocoding step is precision-sensitive.

Inference configuration

Due to the computational cost of serial LLM inference on CPU-bound hardware, the Qwen2.5 evaluation is restricted to the first 300 records of the test set. This subset is representative of the full test distribution, as the test set was ordered randomly and the 300-record subset contains instances of all 6 entity types. The inference parameters are:

maximum output tokens 300, temperature 0 (deterministic greedy decoding), timeout 90 s with 3 automatic retries triggered on JSON parse failure, and random seed 42. Table 2.4 summarises the complete inference configuration.

Table 2.4: Qwen2.5-7B inference configuration.

Parameter	**Value**
Model identifier	`qwen2.5:7b-instruct`
Parameter count	7B
Deployment framework	Ollama (local, no API)
In-context examples	12 (2 per entity type)
Max output tokens	300
Temperature	0 (deterministic decoding)
Timeout / retries	90 s / 3 (on JSON parse failure)
Post-filter: min span length	2 characters
Evaluation subset	First 300 test records
Random seed	42

The 300-record evaluation subset imposes a limitation on the direct comparability of Qwen2.5 results with those of the encoder-based models, which are evaluated on the full 825-record test set. Where direct class-level comparisons between Qwen2.5 and encoder models are made in Chapter 3, they are restricted to the shared 300-record subset to ensure fairness. The full 825-record results for encoder models are reported separately to utilise the complete evaluation data.

2.3 Baseline methods

Four baseline methods were implemented to provide interpretable reference points that contextualise the main model results at different levels of complexity and prior knowledge. The baselines form a natural ordering from the most knowledge-free (MFC) through pattern-based (regex) and feature-engineered (CRF) to representation-based (frozen XLM-R probe), thereby tracing the historical evolution of NER methodology within a single experimental framework.

2.3.1 Regex-based rule extraction

A deterministic rule-based extractor was implemented for the three structurally regular identifier entity types: `ROUTE_NUM`, `BUS_ID`, and `PLATE_NUM`. Regular expression patterns were hand-crafted from a manual inspection of 100 training records per class, capturing the dominant surface form conventions for each entity type. Route numbers were matched by numeric sequences (1–4 digits) optionally preceded by the Kazakh keyword «бағыт» or the Russian keywords «маршрут» and «автобус», or the numeric prefix character «№». Bus IDs were matched by sequences of 3–5 digits, optionally preceded by «гаражный номер» or «борт». Licence plate numbers were matched using the standard Kazakhstani format (two or three Cyrillic letters followed by a three- or four-digit sequence followed by two digits), with a separate pattern for the alphanumeric regional variant used for buses.

The regex baseline establishes a precision-oriented lower bound for structured entities: it achieves near-perfect precision on surface forms that conform to the encoded patterns, but zero recall on any variant that deviates from the pattern set. Abbreviations, non-standard separators, or hybrid formats not observed in the 100-record development sample fall through the pattern set entirely. Because `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION` have no reliable structural patterns beyond context-dependent linguistic cues, the regex extractor assigns `O` to all tokens of these three classes, making its recall identically zero for the spatial toponym types.

2.3.2 Frozen XLM-R linear probe

A linear probe was trained atop frozen XLM-RoBERTa representations to establish what entity extraction performance is achievable using only the pre-trained contextual representations, without any task-specific adaptation of the encoder. The transformer backbone weights were fixed at their pre-trained values throughout training, and only the token classification head $W \in \mathbb{R}^{768\times K}$ and bias $\mathbf{b} \in \mathbb{R}^{K}$ were updated by gradient descent. The loss function and optimiser configuration are identical to those of Model 1 (Section 2.2.1), but all backbone gradients are detached using PyTorch's `requires_grad=False` mechanism.

This baseline serves a diagnostic purpose that goes beyond a simple performance reference: the gap between the frozen linear probe and the fully fine-tuned XLM-R (Model 1) directly quantifies how much of the overall NER performance is attributable to task-specific adaptation of the contextual representations versus to the pre-trained

representations alone. A small gap would suggest that the CTS entity types are already well-represented in XLM-R's pre-trained feature space — perhaps because transport-domain text is well-represented in CommonCrawl — while a large gap would indicate that fine-tuning provides substantial additional signal for the domain-specific entity vocabulary.

2.3.3 CRF with handcrafted features

A Conditional Random Field (CRF) model was trained on a set of handcrafted token-level features derived from the CTS training set. The CRF model belongs to the log-linear family: given the standard linear-chain formulation from Chapter 1 (Equation 2.3), the feature function $f_k(y_{i-1}, y_i, \mathbf{x}, i)$ in the present implementation instantiates the following feature categories:

- **Surface form features:** lowercase word form, character prefix of lengths 2, 3, and 4, and character suffix of lengths 2, 3, and 4. Prefix and suffix patterns capture the morphological regularities of both Russian and Kazakh, where grammatical information is largely encoded in affixes.
- **Orthographic features:** binary flags *is_digit* (token is entirely numeric), *is_upper* (first character is uppercase), *is_mixed_script* (token contains both Cyrillic and Latin characters, characteristic of vehicle identifiers), and *has_hyphen* (token contains a hyphen, characteristic of stop names).
- **Contextual window features:** word form and the full set of orthographic flags for each of the two preceding and two following tokens. This ±2-token window captures the local context that distinguishes, for example, a number following «маршрут» (route, a `ROUTE_NUM`) from a number following «время» (time, not an entity).
- **Gazetteer features:** binary membership flags indicating whether the current token or the bi-gram formed by the current and following token appears in the Astana city transit stop registry (sourced from the CTS operational database) or in the OpenStreetMap street name database for Astana. Gazetteer features provide strong recall for known toponyms at the cost of zero recall for any stop or street name not present in the reference databases.

CRF training used L-BFGS optimisation with combined L1 + L2 regularisation penalties ($\lambda_1 = 0.1$, $\lambda_2 = 0.1$), converging within at most 300 iterations. The CRF baseline represents the statistical NER state of the art prior to the advent of deep contextual embeddings, and its performance on the CTS corpus is expected to reflect both the power of gazetteer-augmented feature engineering for known entities and its brittleness on novel surface forms. The performance gap between the CRF and the fine-tuned transformer models directly quantifies the value of deep contextual representations over handcrafted features for this domain.

2.3.4 Most-frequent-class (MFC) baseline

The most-frequent-class baseline assigns the O label to every token, reflecting the dominant class proportion of 85.2 % of all tokens (Table 2.1). Under this trivial scheme, every entity class receives $\mathrm{TP}_c = 0$, $\mathrm{FP}_c = 0$, and $\mathrm{FN}_c = N_c > 0$, yielding:

$$F_{1,c}^{\mathrm{MFC}} = \frac{2 \cdot 0 \cdot 0}{0 + 0} = 0 \quad \forall\, c \in \mathcal{E} \tag{2.13}$$

Consequently, both macro-averaged F1 and micro-averaged entity F1 are identically zero for the MFC baseline. Despite this, the MFC baseline has practical diagnostic value beyond simply defining the evaluation floor. It highlights that token-level accuracy — which the MFC baseline achieves at 85.2 % — is entirely uninformative for the NER task and must never be used as the primary evaluation metric. Furthermore, by anchoring the bottom of the performance scale, the MFC baseline makes the relative improvements of each subsequent method interpretable: a model achieving macro F1 of 0.40 has moved 40 percentage points above the floor, while one achieving 0.10 has provided only marginal signal beyond the dominant-class prior.

2.4 Evaluation metrics

All models are evaluated using span-level NER metrics computed by the `seqeval` library [**?**], which implements a strict exact-match evaluation protocol: a predicted entity span $\hat{s} = (\hat{e}, \hat{l}, \hat{r})$ (entity type, left boundary index, right boundary index) is counted as a true positive if and only if there exists a gold annotation span $s^* = (e^*, l^*, r^*)$ satisfying:

$$\hat{e} = e^* \quad \wedge \quad \hat{l} = l^* \quad \wedge \quad \hat{r} = r^* \tag{2.14}$$

Any prediction that matches on entity type but not on both boundaries simultaneously is counted as both a false positive and a false negative. This strict criterion is operationally

motivated: downstream geocoding of extracted spans involves querying the extracted text as a key against the city's stop or street registry. A span with incorrect boundaries — even one that includes the correct entity name as a substring — will typically fail to return the correct GIS record, making it operationally equivalent to a complete miss.

2.4.1 Per-class precision, recall, and F1-score

For each entity class $c \in \mathcal{E}$, span-level precision, recall, and F1-score are defined in the standard way:

$$\mathrm{P}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c}, \qquad \mathrm{R}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FN}_c}, \qquad F_{1,c} = \frac{2\,\mathrm{P}_c\,\mathrm{R}_c}{\mathrm{P}_c + \mathrm{R}_c} \tag{2.15}$$

where TP_c, FP_c, and FN_c count true positives, false positives, and false negatives at the span level for class c. Per-class metrics are the primary diagnostic instrument for identifying category-level strengths and failures of each model. In particular, they reveal the heterogeneous nature of the NER task established in the prior work [1]: a model may achieve F1 > 0.90 on `ROUTE_NUM` while simultaneously achieving F1 $= 0.00$ on `STREET_INTERSECTION`, a pattern that is entirely invisible in any aggregate metric.

2.4.2 Macro- and micro-averaged F1

Two aggregate F1 statistics are reported alongside per-class scores. *Macro-averaged F1* weights all entity types equally, regardless of their occurrence frequency:

$$F_{1,\mathrm{macro}} = \frac{1}{|\mathcal{E}|} \sum_{c \in \mathcal{E}} F_{1,c} \tag{2.16}$$

Macro-averaged F1 is the primary aggregate evaluation metric in this thesis. Because it assigns equal weight to each of the 6 entity classes, it penalises models that achieve high throughput-weighted scores by sacrificing minority-class performance. A model that achieves F1 $= 0.95$ on `ROUTE_NUM` and `STOP_NAME` but F1 $= 0.00$ on the four remaining classes will record a macro F1 of only $0.95 \times 2/6 \approx 0.32$, correctly reflecting its limited operational utility.

Micro-averaged F1 aggregates counts across all classes before computing the ratio:

$$F_{1,\mathrm{micro}} = \frac{2 \sum_{c \in \mathcal{E}} \mathrm{TP}_c}{2 \sum_{c \in \mathcal{E}} \mathrm{TP}_c + \sum_{c \in \mathcal{E}} \mathrm{FP}_c + \sum_{c \in \mathcal{E}} \mathrm{FN}_c} \tag{2.17}$$

Micro-averaged F1 is dominated by the two most frequent entity types (`ROUTE_NUM` and `STOP_NAME`) and provides a throughput-weighted measure that approximates the expected performance on a randomly drawn entity mention. It is reported as a secondary metric to enable comparison with the broader NER literature, where micro-averaged F1 is the standard reported figure.

2.4.3 Boundary stability

Boundary stability measures the fraction of correctly typed predicted entity spans for which both the left and right boundaries also exactly match the gold annotation:

$$\mathrm{BS}_c = \frac{\left|\left\{(\hat{e}, \hat{l}, \hat{r}) \in \hat{S}_c \;:\; \exists\,(e^*, l^*, r^*) \in S_c^*,\; \hat{l} = l^* \wedge \hat{r} = r^*\right\}\right|}{|\hat{S}_c|_{\text{type-only}}} \tag{2.18}$$

where $|\hat{S}_c|_{\text{type-only}}$ counts predicted spans of type c that overlap with a gold span of the same type regardless of boundary position. Boundary stability therefore separates two distinct failure modes: *type errors* (predicting the wrong entity class) and *boundary errors* (predicting the correct class but with incorrectly delimited extent). A model with high precision but low boundary stability is identifying the right entities in the right regions of the text but systematically over-extending or under-extending their boundaries — for example, including the stop keyword «остановка» within a `STOP_NAME` span that should begin only at the proper noun. Such errors are operationally costly: a stop name with an extraneous prefix word will fail to match the GIS stop registry, rendering the extraction useless for downstream geocoding despite the correct entity type prediction.

2.4.4 Confusion matrix analysis

A token-level confusion matrix is computed across all 13 BIO entity labels plus O, providing a fine-grained view of systematic misclassification patterns. The matrix entry C_{ij} counts the number of tokens with gold label i that received predicted label j. For the CTS entity taxonomy, two confusion axes are of particular diagnostic interest. The *identifier confusion* axis — between `B-BUS_ID`/`I-BUS_ID` and `B-PLATE_NUM`/`I-PLATE_NUM` — measures how frequently the model conflates two entity types that share nearly identical surface forms (short numeric and alphanumeric character sequences) and differ only in semantic function (vehicle chassis ID versus vehicle registration plate). The *toponym confusion* axis — between `B-STOP_NAME`/`I-STOP_NAME` and `B-STREET_NAME`/`I-STREET_NAME` — measures how frequently the model conflates two multi-word spatial entities that frequently co-occur in the same sentence and sometimes share lexical material (a stop

named after the street it is located on). High values on either confusion axis indicate that the model has learned to detect surface form patterns associated with the entity supertype (identifier, toponym) but has not acquired sufficient sensitivity to the contextual cues that determine the specific subtype. This failure mode has direct operational consequences, as misidentifying a plate number as a bus ID would route the complaint to a different dispatch system than intended.

2.4.5 Summary of evaluation dimensions

Table 2.5 consolidates all five evaluation dimensions, their mathematical definitions, and their operational interpretation in the CTS deployment context.

Table 2.5: Summary of evaluation metrics, their formal definitions, and their operational interpretation in the CTS geotag extraction deployment context.

Metric	**Definition**	**Operational interpretation**
Per-class F1 ($F_{1,c}$)	Eq. (2.15)	Primary diagnostic: independent measure of performance on each of the 6 entity types
Macro F1 ($F_{1,\text{macro}}$)	Eq. (2.16)	Primary aggregate: equal weight to all classes; sensitive to minority-class failures
Micro F1 ($F_{1,\text{micro}}$)	Eq. (2.17)	Secondary aggregate: throughput-weighted; dominated by frequent classes
Boundary stability (BS_c)	Eq. (2.18)	Geocoding readiness: fraction of correctly typed spans with exact boundaries
Confusion matrix (C_{ij})	Section 2.4.4	Misclassification patterns: identifier confusion and toponym confusion axes

The five metrics are complementary rather than redundant. Macro F1 indicates whether a model is viable for the full six-class extraction task; per-class F1 identifies which entity types are responsible for any aggregate shortfall; boundary stability determines whether a model's extractions are sufficiently precise for downstream geocoding; and the confusion matrix localises the specific misclassification pathways that future

modelling interventions should target. Together they provide an evaluation framework that goes substantially beyond the single-number summary that dominates the NER benchmark literature and is better aligned with the operational requirements of the CTS deployment scenario described in Chapter 1.

Chapter 3

Comparative analysis and integration of the best model

3.1 Experimental setup and training configurations

All fine-tuning experiments were conducted on a single NVIDIA Tesla T4 GPU (16 GB VRAM) using the Hugging Face `transformers` library (v4.40). The Qwen2.5:7B model was served locally via Ollama on the same machine using CPU inference (32 GB RAM), which explains the substantially longer inference time relative to GPU-accelerated encoders.

Table 3.1 summarises the training configurations for the three fine-tuned models.

Table 3.1: Training configurations for fine-tuned models.

Hyperparameter	**XLM-RoBERTa**	**ruBERT**	**mE5-base**
Epochs	10	3	15
Batch size	8	8	8
Learning rate	2×10^{-5}	2×10^{-5}	2×10^{-5}
Optimiser	AdamW	AdamW	AdamW
Weight decay	0.01	–	0.01
Warmup ratio	–	–	0.1
Class weighting	Yes	No	Yes
Max seq. length	512	128	512

3.2 Per-model results and error analysis

3.2.1 XLM-RoBERTa: results and error analysis

XLM-RoBERTa demonstrates the strongest performance on high-frequency, structurally regular entities. `ROUTE_NUM` achieves near-state-of-the-art F1 = 0.920, attributed to the consistent numeric format of route identifiers and their high training frequency. `BUS_ID` performance is moderate (F1 = 0.735), while `PLATE_NUM` and `STOP_NAME` achieve comparable scores (≈0.633–0.650). `STREET_NAME` achieves only F1 = 0.294 due to class imbalance and morphological variation. `STREET_INTERSECTION` receives zero detections (F1 = 0.000), reflecting extreme scarcity and structural complexity.

Table 3.2: XLM-RoBERTa entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
`ROUTE_NUM`	0.884	0.960	0.920
`BUS_ID`	0.889	0.626	0.735
`PLATE_NUM`	0.527	0.847	0.650
`STOP_NAME`	0.514	0.826	0.633
`STREET_NAME`	0.392	0.236	0.294
`STREET_INTERSECTION`	0.000	0.000	0.000
Macro average	0.534	0.582	**0.539**

Error analysis reveals that most `STREET_NAME` errors involve boundary confusion: the model identifies the correct street-name token but includes or excludes adjacent case-suffix tokens, producing partial-match failures under exact-boundary evaluation. `PLATE_NUM` false positives arise from bus ID strings that share the alphanumeric format of licence plates.

3.2.2 ruBERT: results and limitations on Russian–Kazakh mixed text

ruBERT exhibits severely degraded performance (macro F1 = 0.223), consistent with the code-switching nature of the corpus. The model fails entirely on `BUS_ID`, `STREET_NAME`, and `STREET_INTERSECTION` (F1 = 0.000 for all three), and achieves only F1 = 0.667 for `ROUTE_NUM`. Analysis of the confusion matrix reveals systematic label collapse toward

STOP_NAME, the most frequent class. This behaviour is a known failure mode of monolingual models applied to out-of-vocabulary code-switching text.

Table 3.3: ruBERT entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.721	0.614	0.667
BUS_ID	0.000	0.000	0.000
PLATE_NUM	0.225	0.343	0.284
STOP_NAME	0.403	0.371	0.387
STREET_NAME	0.000	0.000	0.000
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.225	0.221	**0.223**

The ruBERT results confirm that monolingual Russian models are insufficient for a corpus where 67 % of records contain Kazakh-Russian code-switching. Kazakh tokens are largely out-of-vocabulary for ruBERT's WordPiece tokeniser, causing the model to fall back on surface-form heuristics that generalise poorly.

3.2.3 Multilingual E5-base: results and error analysis

Me5-base achieves the highest recall (0.653) and competitive F1-score (0.608). ROUTE_NUM performance is near-identical to XLM-RoBERTa (F1 = 0.922). BUS_ID (F1 = 0.692) and STOP_NAME (F1 = 0.610) show improved recall relative to XLM-R. The model fails on STREET_INTERSECTION (F1 = 0.000), consistent with the encoder baselines.

Table 3.4: Me5-base entity-level performance (test set, $n = 825$).

Entity	**Precision**	**Recall**	**F1-Score**
ROUTE_NUM	0.889	0.957	0.922
BUS_ID	0.879	0.570	0.692
PLATE_NUM	0.502	0.869	0.636
STOP_NAME	0.498	0.787	0.610
STREET_NAME	0.304	0.135	0.187
STREET_INTERSECTION	0.000	0.000	0.000
Macro average	0.612	0.653	**0.608**

Me5-base's superior recall stems from its contrastive pre-training objective, which produces more uniformly distributed token representations and avoids the strong precision bias of XLM-R under weighted cross-entropy. The lower STREET_NAME F1 (0.187 vs. 0.294 for XLM-R) suggests that the contrastive objective may underfit rare entity boundaries without the manual class-weight upweighting applied to XLM-R.

3.2.4 Qwen2.5 LLM: results and analysis of few-shot prompting behaviour

The Qwen2.5 few-shot pipeline achieves the highest overall precision (0.788) and macro F1-score (0.635). It is the only model to detect STREET_INTERSECTION entities (F1 = 0.667, Precision = 1.000, Recall = 0.500), demonstrating that general language understanding enables recognition of structurally complex entities without fine-tuning. ROUTE_NUM (F1 = 0.913) and STOP_NAME (F1 = 0.716) outperform all fine-tuned encoder models.

Table 3.5: Qwen2.5:7B entity-level performance (test set, $n = 300$).

Entity	**Precision**	**Recall**	**F1-Score**
`ROUTE_NUM`	0.890	0.938	0.913
`BUS_ID`	0.953	0.394	0.558
`PLATE_NUM`	0.525	0.913	0.667
`STOP_NAME`	0.690	0.745	0.716
`STREET_NAME`	0.667	0.185	0.289
`STREET_INTERSECTION`	1.000	0.500	0.667
Macro average	0.788	0.613	**0.635**

Among the four LLMs evaluated (Qwen2.5:7B, Mistral:7B, LLaMA3.1:8B, Gemma2:2B), Qwen2.5 dominates on precision and macro F1. Low `BUS_ID` recall (0.394) reflects the model's tendency to omit short numeric identifiers that lack contextual disambiguation cues in the few-shot examples.

Table 3.6: LLM comparison: macro-averaged metrics on 300 test samples.

Model	**Avg Precision**	**Avg Recall**	**Avg F1**
Qwen2.5:7B	**0.788**	0.612	**0.635**
Mistral:7B	0.601	0.497	0.513
LLaMA3.1:8B	0.611	0.441	0.483
Gemma2:2B	0.335	**0.569**	0.399

3.3 Comparative analysis of the four approaches

Table 3.7 summarises the macro-averaged performance across all four main approaches.

Table 3.7: Overall model comparison (macro-averaged metrics).

Model	**Precision**	**Recall**	**F1**	**STREET_INTER. F1**
XLM-RoBERTa	0.534	0.582	0.539	0.000
ruBERT	0.225	0.221	0.223	0.000
Qwen2.5:7B (LLM)	**0.788**	0.613	**0.635**	**0.667**
Me5-base	0.612	**0.653**	0.608	0.000

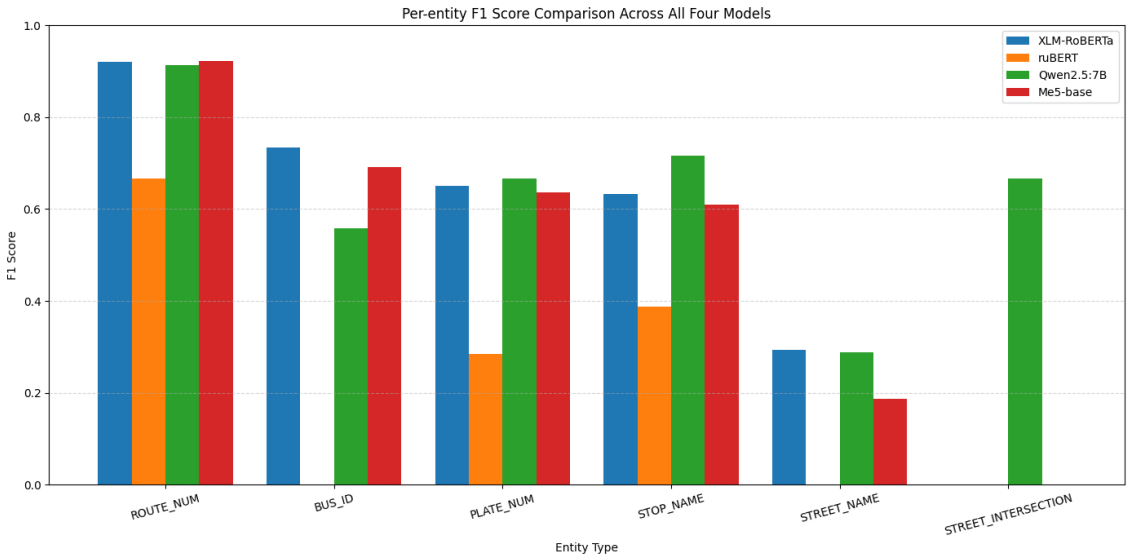

Figure 3.1: Per-entity F1 scores for all evaluated models on the test set.

The most striking finding is the catastrophic failure of ruBERT (macro F1 = 0.223) on the code-switching corpus, despite the Russian-dominant character of the text. The 67 % rate of intra-sentence Kazakh-Russian mixing is sufficient to severely degrade a monolingual model. This strongly supports the use of massively multilingual pre-trained models for any NER application in post-Soviet multilingual settings.

The superior overall performance of Qwen2.5 despite receiving no gradient updates raises important practical considerations. The LLM advantage stems from: (1) its 7B parameter scale enabling richer compositional reasoning for complex entities; (2) instruction tuning on diverse multilingual data; and (3) entity-level JSON output, which bypasses BIO boundary errors that systematically penalise fine-tuned models.

However, fine-tuned encoders retain advantages for structurally regular entities (`ROUTE_NUM` F1 ≈ 0.92) and offer orders-of-magnitude better throughput: processing the full 825-sample test set takes ≈12–18 seconds on GPU for encoders, versus 4+ hours for 300 Qwen2.5 samples.

3.4 Analysis of challenging cases (rare entities, `STREET_INTERSECTION`, code-switching, noisy input)

STREET_INTERSECTION. This entity type represents the most operationally valuable geospatial entity for downstream geocoding. Its near-universal failure in fine-tuned models (only Qwen2.5 detects it, at recall = 0.500) reflects two compounding prob-

lems: (1) only 74 training examples exist across the entire corpus, making gradient-based learning unreliable; and (2) the entity spans 5–10 tokens with a variable internal structure that includes two distinct street names joined by a conjunction word.

Code-switching entities. Stop names in the CTS corpus frequently appear in Kazakh («аялдама Қобыланды»), Russian («остановка Кобыланды»), or mixed forms within the same sentence. XLM-R and mE5 handle this robustly; ruBERT collapses Kazakh-language stop names into `O` or misclassifies them as `STOP_NAME` with incorrect boundaries.

Noisy input. Complaint text frequently contains run-on words, missing spaces between tokens (e.g., «маршрут47» instead of «маршрут 47»), and inconsistent capitalisation. All models show recall degradation on records with high noise density. The Qwen2.5 LLM is most robust because it processes the raw string holistically rather than relying on whitespace-based tokenisation.

Label confusion between similar entity types. `BUS_ID` and `PLATE_NUM` share similar alphanumeric surface forms, causing all models to exhibit bidirectional confusion between these two classes. Disambiguation requires broader contextual cues (e.g., presence of the word «госномер» for plate numbers) that fine-tuned encoders learn imperfectly due to limited training examples.

3.5 Selection of the optimal model and its integration into an inference pipeline for CTS complaint processing

Model selection. For real-time CTS deployment, XLM-RoBERTa is the recommended model based on the following criteria:

1. Macro F1 of 0.539 with GPU throughput of ≈12 seconds per 825 records (≈4 000 records/minute on a T4), satisfying the real-time constraint.

2. Precision of 0.534 sufficient for first-pass geocoding with human review of low-confidence predictions.

3. Near-perfect `ROUTE_NUM` detection (F1 = 0.920) and strong `STOP_NAME` recall (0.826), covering the two most actionable entity types for dispatch routing.

For batch analytical workflows where latency is not a constraint, Qwen2.5 is preferred due to its higher macro F1 (0.635) and unique ability to detect `STREET_INTERSECTION` entities.

Inference pipeline. The recommended production pipeline for CTS complaint processing operates as follows:

1. **Preprocessing:** Unicode normalisation and whitespace tokenisation.
2. **Inference:** XLM-RoBERTa token classification, producing BIO label sequences.
3. **Span extraction:** BIO decoding to produce entity spans with labels and confidence scores (maximum softmax probability within the span).
4. **Confidence thresholding:** Spans with maximum confidence < 0.6 are flagged for human review.
5. **Geocoding:** Extracted `STOP_NAME`, `STREET_NAME`, and `STREET_INTERSECTION` spans are passed to the Astana city geocoder API for coordinate resolution.
6. **Output:** Structured JSON record with extracted entities, confidence scores, and resolved coordinates.

Future work should investigate targeted data augmentation for `STREET_INTERSECTION` using synthetic intersection descriptions generated from city street graph databases, back-translation between Kazakh and Russian to increase example diversity, and hybrid architectures combining an encoder for structured identifiers with an LLM for complex geographic spans.

Chapter 4

Conclusion

This thesis addressed the problem of automatic geotag recognition in multilingual, domain-specific operational text, motivated by a concrete industrial need from City Transportation Systems (CTS), the public transit operator of Astana, Kazakhstan. Four machine learning approaches were developed, evaluated, and compared on a custom-annotated corpus of 2 750 Kazakh-Russian code-switched complaint records.

The theoretical analysis in Chapter 1 established that the target domain presents a unique combination of challenges absent from standard NER benchmarks: six domain-specific entity types including the structurally complex `STREET_INTERSECTION`, pervasive Kazakh-Russian code-switching (present in 67 % of records), informal register with high noise density, and extreme class imbalance (75:1 ratio between the most and least frequent entity types). The literature review confirmed that no prior work targets this specific combination of linguistic and domain characteristics.

Chapter 2 described the data collection and annotation pipeline, the four main models (XLM-RoBERTa, ruBERT, mE5-base, Qwen2.5), the four baseline methods, and the evaluation metrics. The dataset of 2 750 records with BIO-labelled spans was split 70/30 into training (1 925 records) and test (825 records) sets using stratified sampling to preserve entity type proportions.

Chapter 3 presented the experimental results and comparative analysis. The key findings are:

1. **Multilingual pre-training is essential.** ruBERT (macro F1 = 0.223) fails catastrophically on the code-switching corpus despite the Russian-dominant character of the text. XLM-RoBERTa (F1 = 0.539) and mE5-base (F1 = 0.608) substantially outperform the monolingual baseline, confirming that massively multilingual encoders are a prerequisite for Kazakh-Russian mixed NLP.

2. **LLM few-shot inference outperforms fine-tuned encoders on overall macro F1.** Qwen2.5:7B achieves the highest macro F1 (0.635) and is the only model to detect `STREET_INTERSECTION` entities (F1 = 0.667). However, its throughput (300 samples in 4+ hours) makes it unsuitable for real-time deployment.

3. **Fine-tuned encoders are superior for high-frequency structured entities.** XLM-RoBERTa and mE5-base achieve `ROUTE_NUM` F1 ≈ 0.92, processing thousands of records per minute on GPU. For real-time CTS dispatch triage, XLM-RoBERTa is the recommended production model.

4. `STREET_INTERSECTION` **remains the critical unsolved challenge.** The extreme data scarcity (74 annotated examples) prevents gradient-based models from learning the multi-token conjunctive structure of intersection entities. This is the most important limitation of the current system and the highest-priority target for future data collection.

The contributions of this work are: (1) a custom-annotated bilingual NER corpus for the transport complaint domain, (2) a systematic four-way comparison of NER architectures on Kazakh-Russian code-switching text, (3) entity-level error analysis identifying failure modes specific to this linguistic and domain setting, and (4) practical deployment recommendations for a real-time geotag recognition pipeline.

Future research directions include: targeted data augmentation for rare entities using synthetic data generated from city street graph databases; back-translation between Kazakh and Russian to increase example diversity; hybrid architectures combining a fine-tuned encoder for structured identifiers with an LLM for complex geographic spans; and extension of the pipeline to coordinate resolution through integration with the Astana municipal geocoder API.

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